

PCB Design Using KiCad

From schematic to fabrication — a hands-on engineering workflow

Visiting Faculty Lecture • Engineering College Programme

Audience: Beginner → Intermediate | Industry-oriented

Open-source EDA • Schematic + Layout + 3D + Gerber • Manufacturing-ready

Introduction to PCB Design

What a PCB is, why it matters, where it shows up

PCBs — From Concept to Critical Component

What	Why	Types	Where
Printed Circuit Board Mechanical substrate + conductive copper paths that connect components.	Replaces hand-wired circuits. Reliable, mass-producible, compact, repeatable.	• Single-sided • Double-sided • Multilayer (4-32) • Flex / Rigid-flex • HDI / Microvia	Phones, laptops, EVs, IoT, medical, aerospace, robotics, automotive, industrial automation.

What is KiCad?

The open-source EDA suite that runs production hardware

Open-source Electronic Design Automation

- ▶ Free, libre, and cross-platform (Windows / macOS / Linux)
- ▶ Maintained by CERN, the KiCad Services Corp., and a global community
- ▶ Production-grade — used by startups, labs, and Fortune-500 hardware teams
- ▶ No node/seat limits, unlimited layers, unlimited pin counts
- ▶ Native Python scripting & a plugin ecosystem
- ▶ Full toolchain: schematic, PCB, 3D, simulation, fabrication output

Quick Facts	
First release	1992
Latest stable	KiCad 8.x / 9.x
License	GPL v3
Languages	20+ localizations
File format	Plain text (Git-friendly)
Cost	₹0 — forever

Features of KiCad

Everything you need under one roof

KiCad — Core Features

Eeschema

Schematic capture
with hierarchical sheets

Symbol Editor

Build & customize
component symbols

PcbNew

Powerful PCB layout
& interactive router

Footprint Editor

Create & manage
landpatterns

3D Viewer

Realistic 3D render
for mechanical fit

Gerber Viewer

Inspect manufacturing
outputs

SPICE Sim

Built-in ngspice
simulation

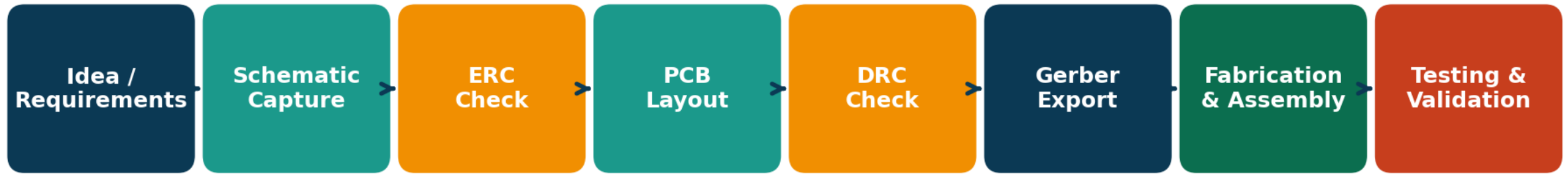
Python API

Scripting,
plugins, automation

PCB Design Workflow

The eight-step flow every engineer follows

End-to-End PCB Design Workflow

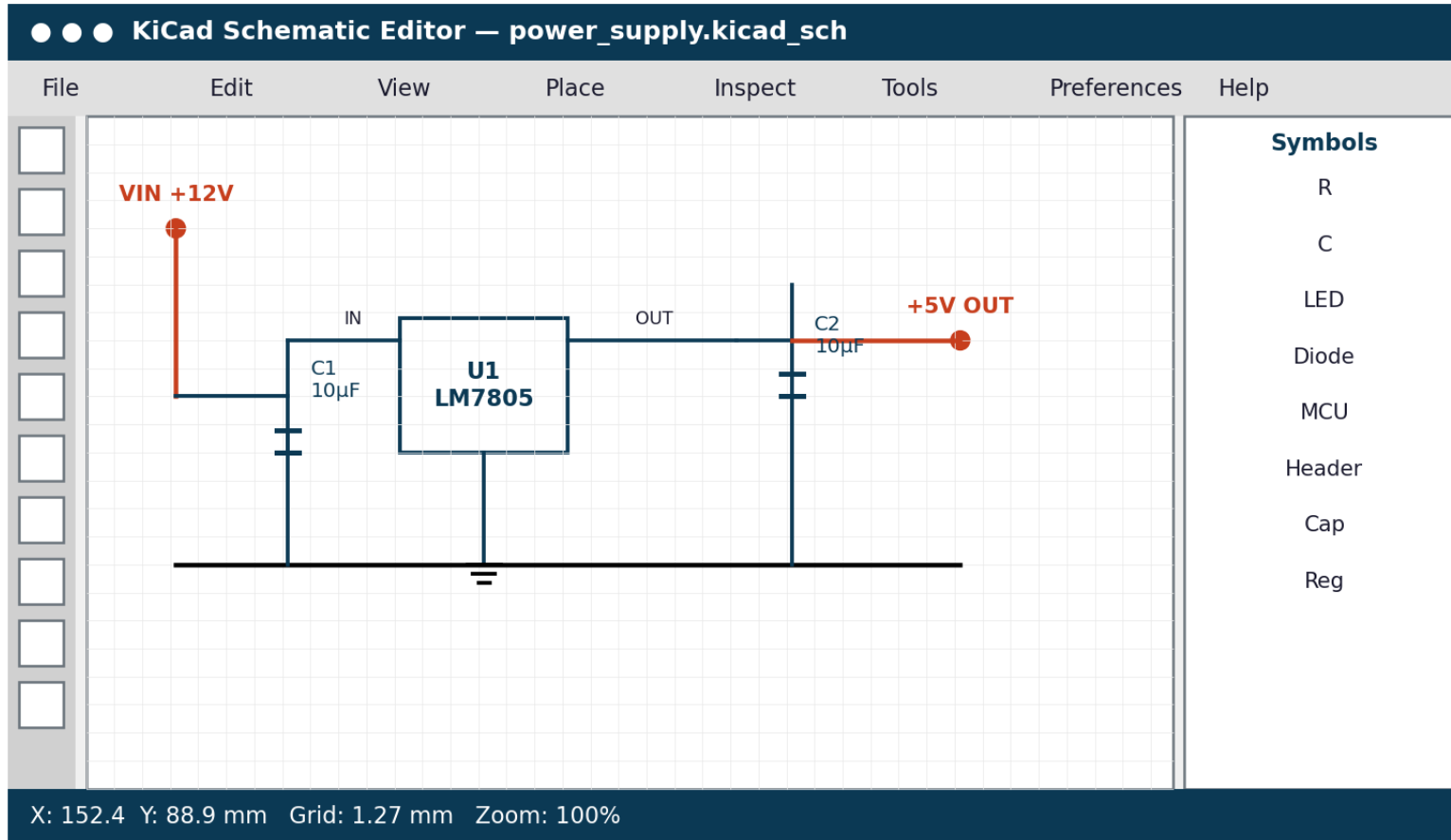


Iterative loop: failed ERC/DRC sends the design back to schematic/layout

Each ERC/DRC failure loops you back — design is iterative, not linear.

Electronic Schematic Design

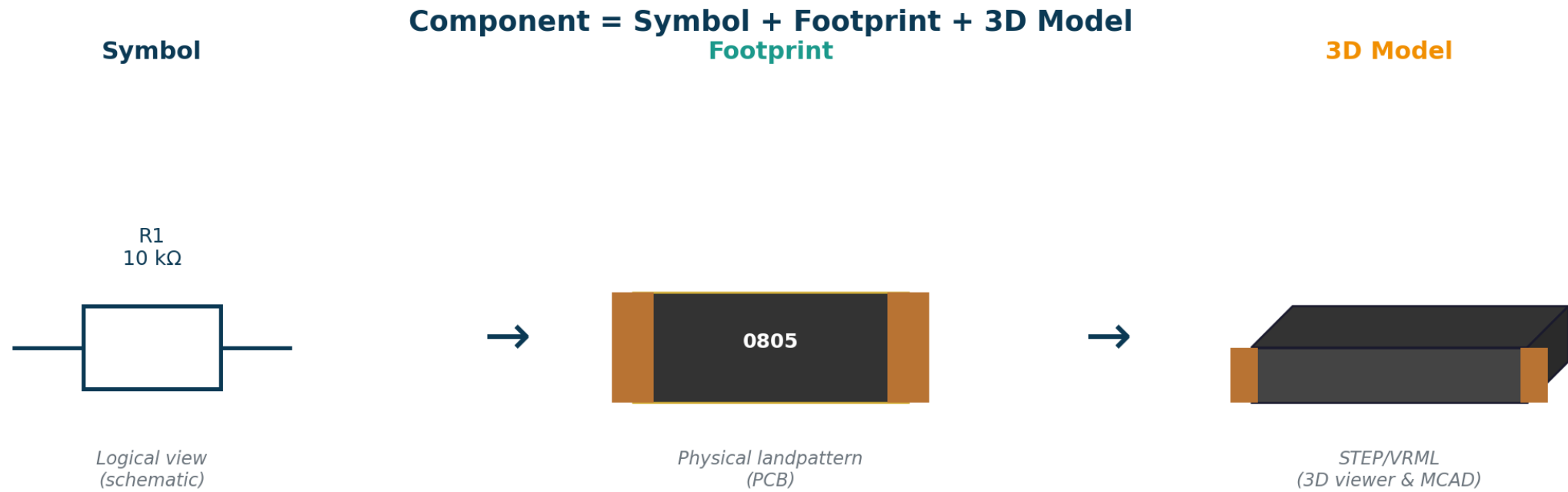
Eeschema — capturing intent before geometry



- ▶ Symbols represent function — not physical shape
- ▶ Wires define electrical connectivity (nets)
- ▶ Power flags mark VCC / GND for ERC
- ▶ Hierarchical sheets for large designs
- ▶ Annotation: R1, C1, U1 ... auto-assigned
- ▶ BOM export per ref-des & value
- ▶ Cross-probes to PCB editor

Component Libraries & Footprints

Symbol + Footprint + 3D model = a usable part



Where do parts come from?

- KiCad built-in library
- SnapEDA / Ultra Librarian / Component Search Engine
- Manufacturer-supplied symbols
- Your own custom library (always verify against the datasheet!)

ERC and DRC Concepts

Automated checks that catch issues before fabrication

Automated Design Checks

ERC — Electrical Rules Check (Schematic)

- ✓ Unconnected pins
- ✓ Conflicting drivers
- ✓ Missing power flags
- ✓ Output → Output
- ✓ Unused power inputs
- ✓ Hierarchical net mismatches

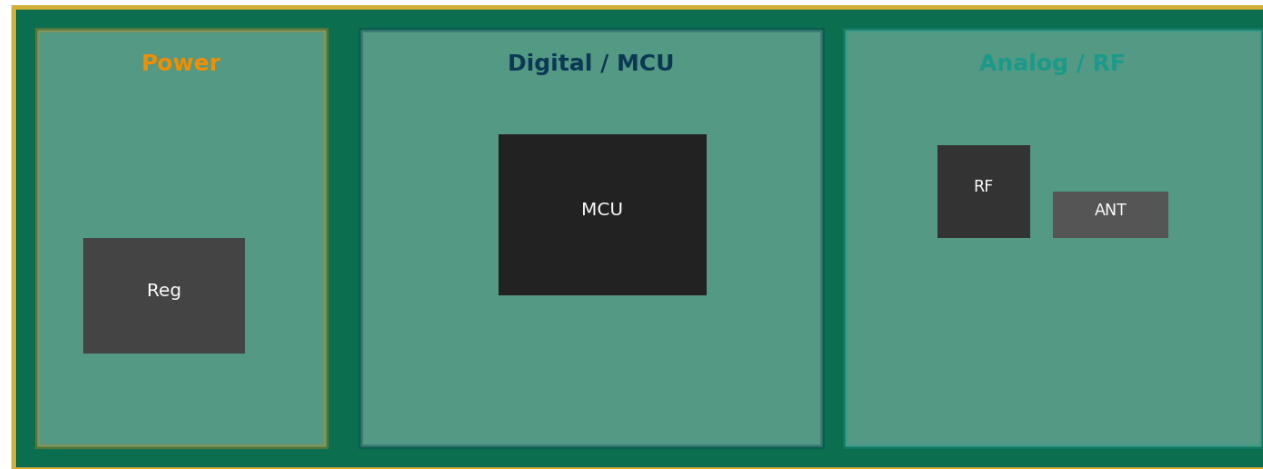
DRC — Design Rules Check (PCB)

- ✓ Trace width violation
- ✓ Clearance violation
- ✓ Hole-to-hole spacing
- ✓ Silk-on-pad
- ✓ Unrouted nets
- ✓ Courtyard overlaps

PCB Layout Fundamentals

Placement before routing — 80% of the work, 20% of the time

Component Placement Fundamentals



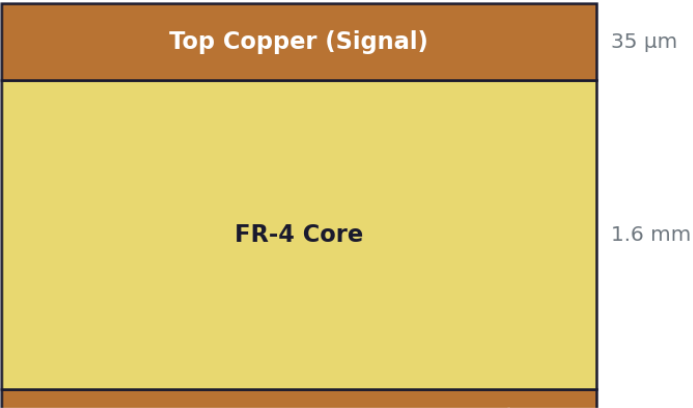
- 1. Group components by function
- 2. Place connectors at board edges
- 3. Keep high-speed traces short
- 4. Place decoupling caps near IC pins
- 5. Separate analog & digital sections
- 6. Allow space for thermal & mounting

Layer Stackup

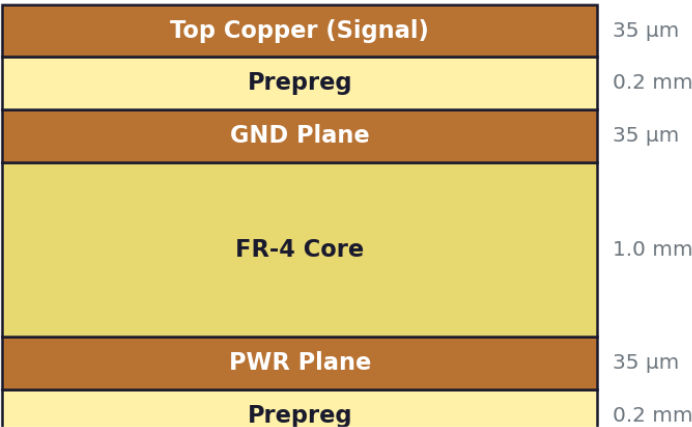
Choosing 2-layer vs 4-layer vs higher — a cost/performance trade

PCB Layer Stackup Comparison

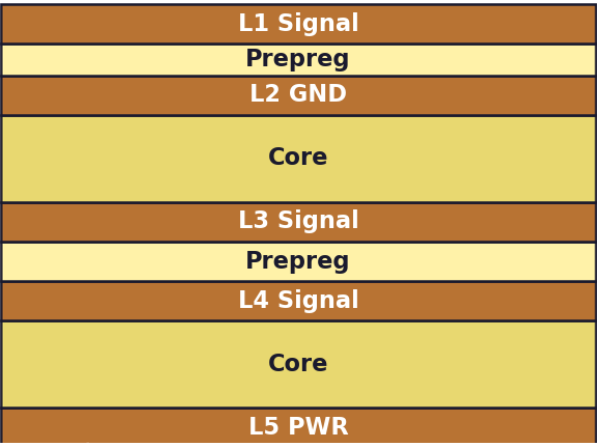
2-Layer Board



4-Layer Board



6-Layer Board



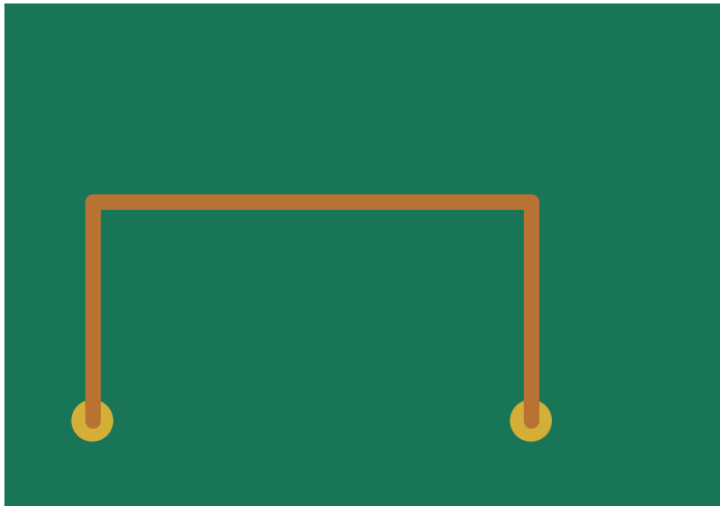
	2-layer	4-layer	6-layer+
Cost (proto, 10 pcs)	₹500–1500	₹2000–5000	₹8000+
Signal integrity	Limited	Good	Excellent
EMI performance	Poor	Good	Best
Best for	Hobby, low-speed	Most products	High-speed, RF, dense

Routing Techniques

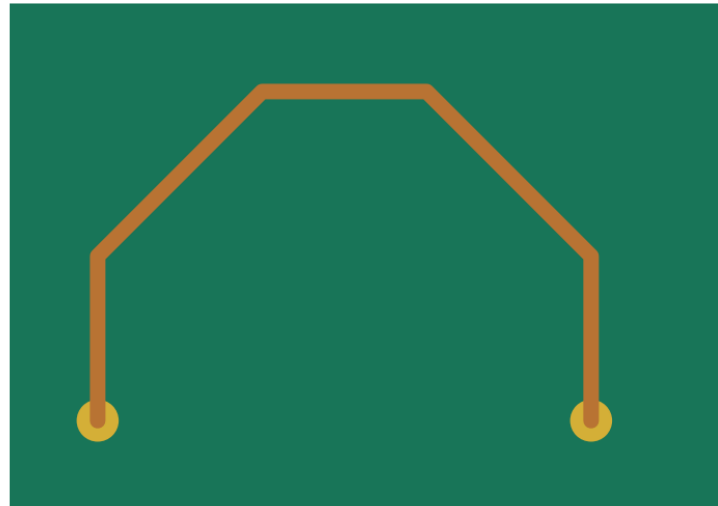
How traces are placed determines signal quality

Routing Corner Techniques

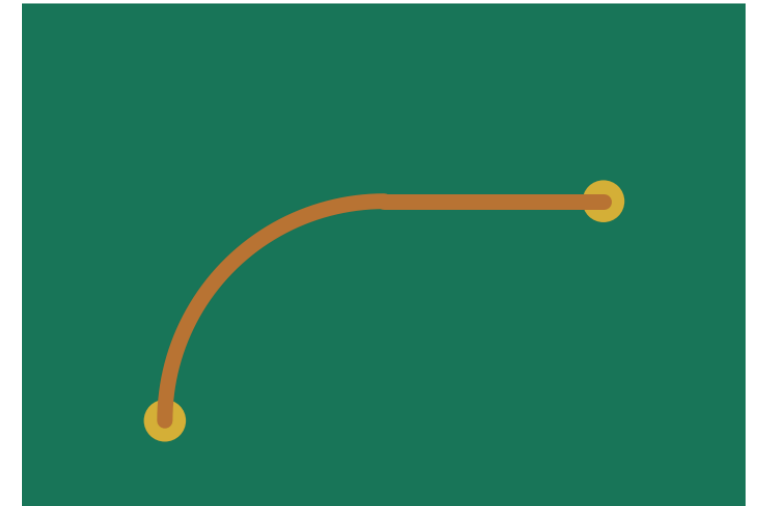
90° corners (avoid)



45° corners (good)



Curved (best for HF)



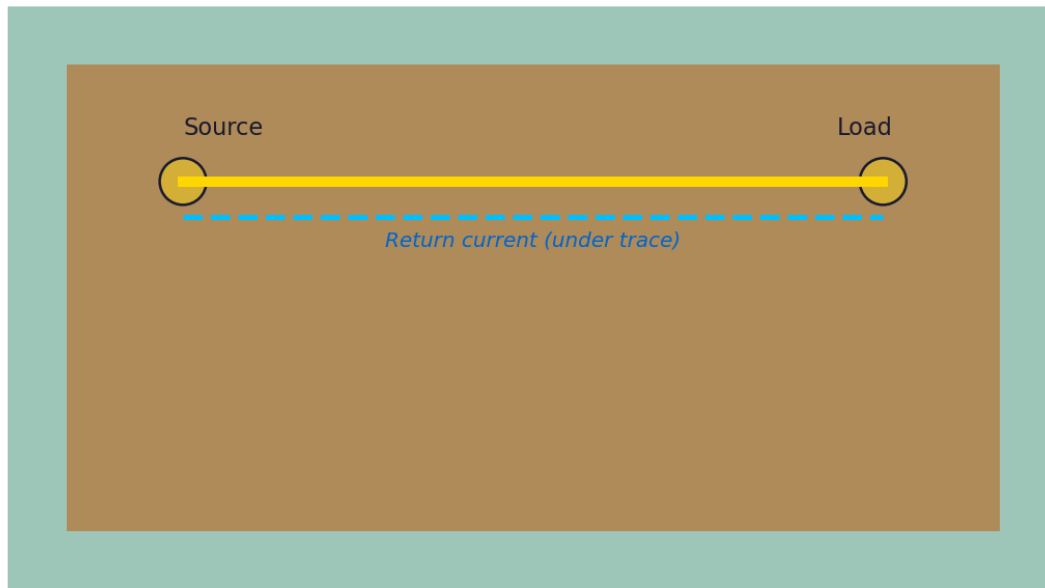
- ▶ Use 45° corners — 90° causes acid traps and impedance discontinuities
- ▶ Differential pairs: route as a pair, length-matched within 5 mil (0.13 mm)
- ▶ Keep high-speed traces on the layer adjacent to a continuous ground plane
- ▶ Power traces sized for current — 10 mil $\approx$ 1 A on 1-oz copper at 10 °C rise

Ground Plane & Power Distribution

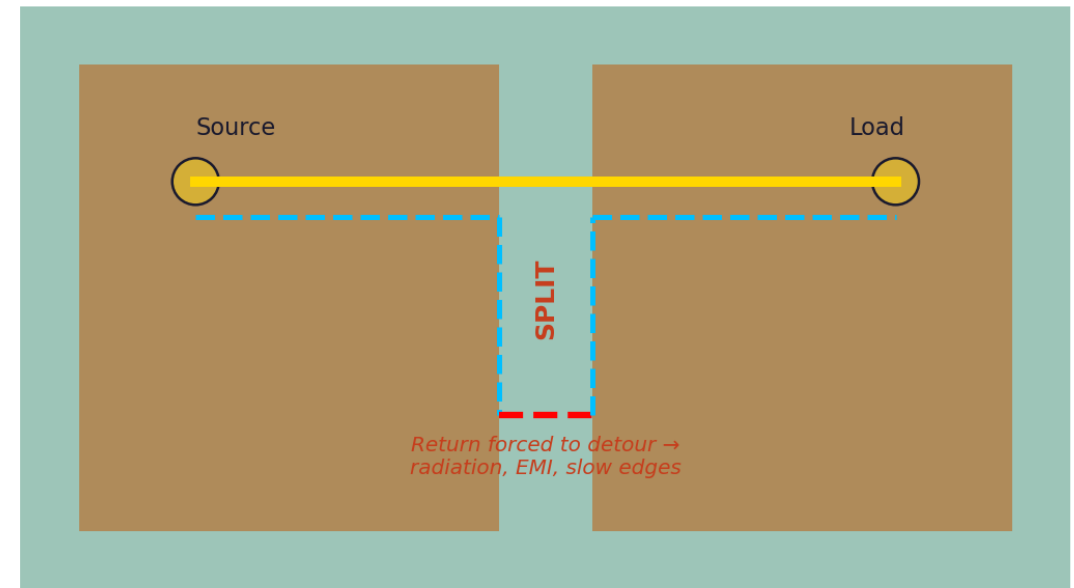
Return paths matter as much as forward paths

Ground Plane: Why Continuity Matters

Solid Ground Plane (Good)



Split Ground Plane (Bad)

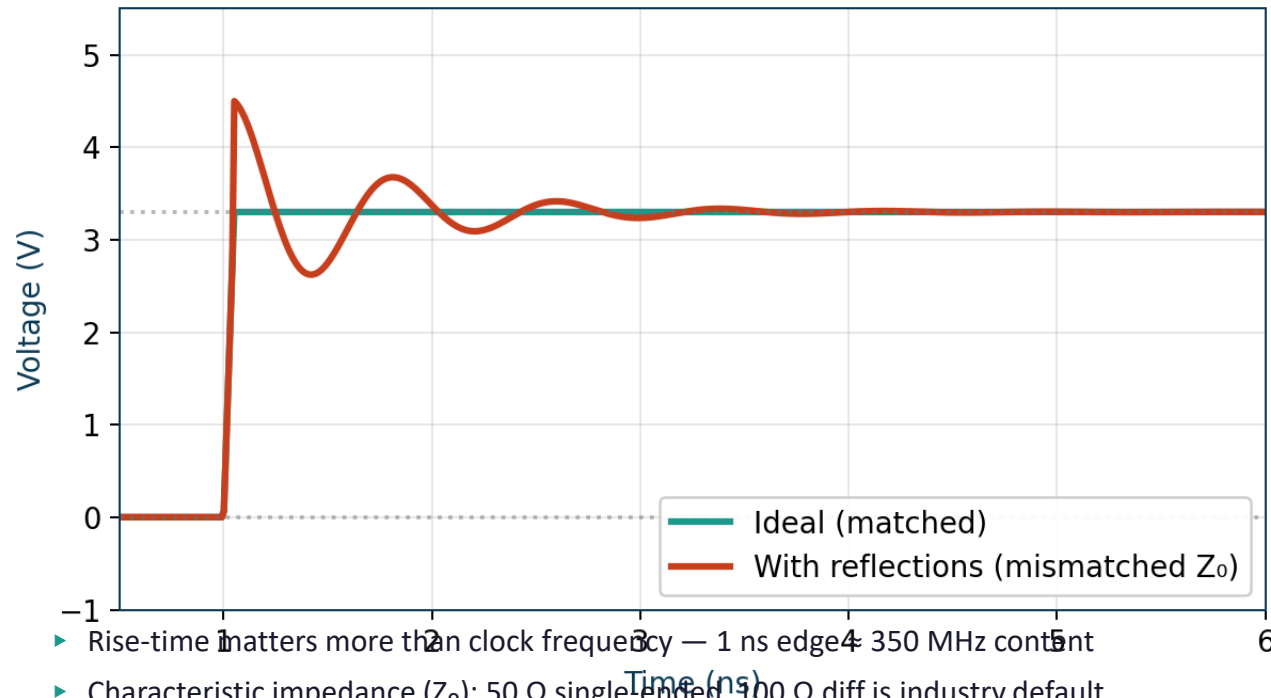


Signal Integrity Basics

When wires stop behaving like wires

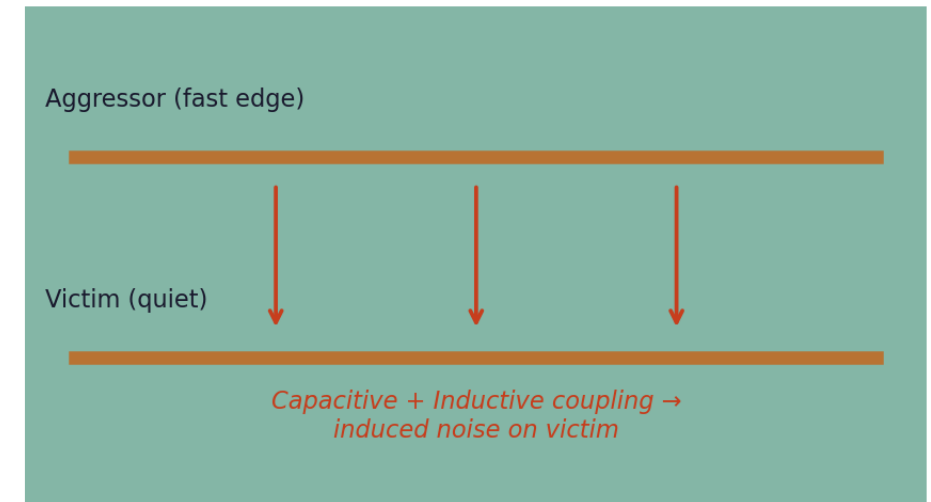
Signal Integrity Phenomena

Reflections & Ringing



- ▶ Rise-time matters more than clock frequency — 1 ns edge = 350 MHz content
- ▶ Characteristic impedance (Z_0): 50 Ω single ended, 100 Ω diff is industry default
- ▶ Termination tames reflections — series, parallel, or AC depending on topology
- ▶ Crosstalk falls quickly with spacing — keep $\geq 3\times$ trace width between victim & aggressor

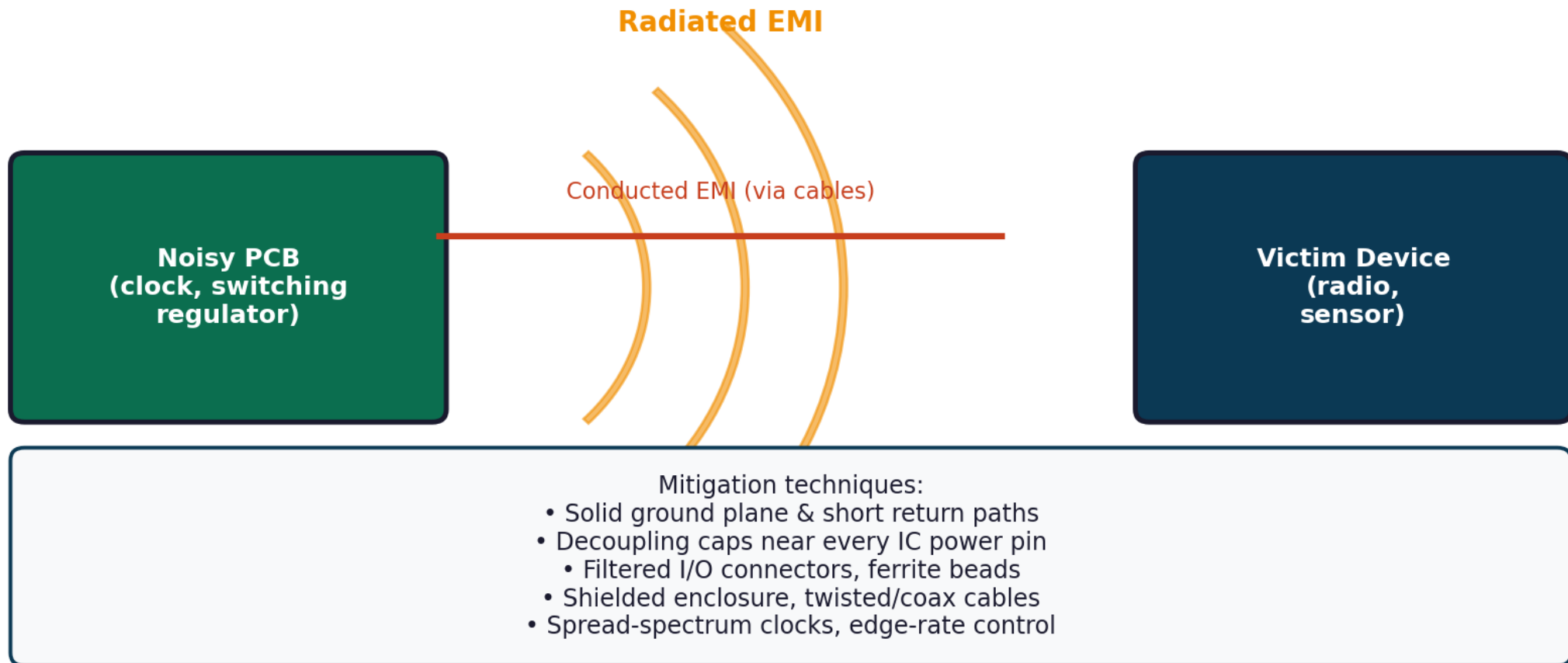
Crosstalk Between Adjacent Traces



EMI / EMC Considerations

Pass the chamber on the first try

EMI / EMC Concerns in PCB Design

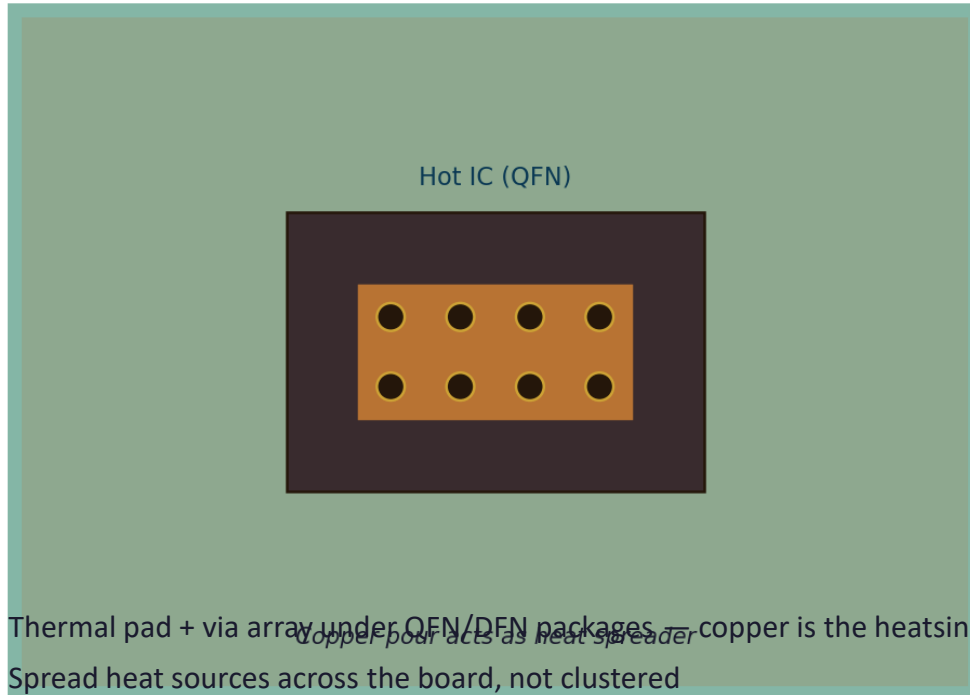


Thermal Management

Heat is the enemy of reliability

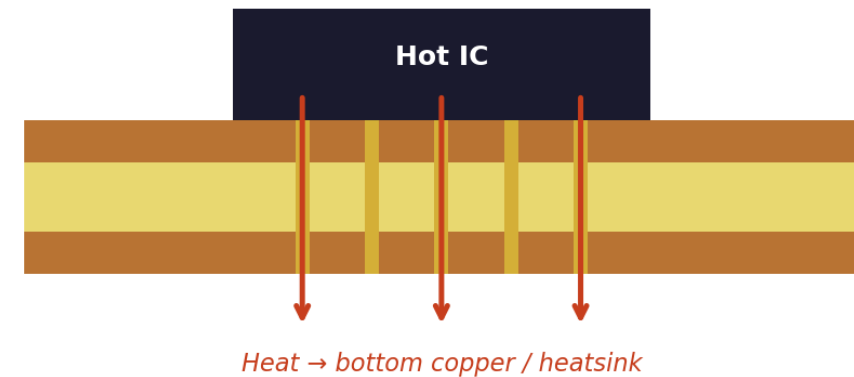
Thermal Management on PCB

Top View: Thermal Pad + Via Stitching



- ▶ Thermal pad + via array under QFN/DFN packages — copper is the heatsink
- ▶ Spread heat sources across the board, not clustered
- ▶ Each 10 °C rise halves component lifetime (Arrhenius)
- ▶ Calculate: $P \times \theta_{JA} = \Delta T \rightarrow$ size copper area accordingly

Side View: Heat Flow Path



Design for Manufacturability (DFM)

Designing within your fab's capabilities

Design for Manufacturability — Key Constraints

Trace width ≥ 0.15 mm

Standard fabs ≥ 6 mil

Annular ring ≥ 0.15 mm

Pad – hole radius

Min via drill 0.3 mm

Smaller = HDI cost

Min clearance 0.15 mm

Trace-to-trace/pad

Silkscreen ≥ 0.8 mm text

Readable after print

Solder mask 0.1 mm gap

Prevents bridging

Edge clearance 0.3 mm

Routing/V-cut safety

Standard panel sizes

Cheaper fabrication

Gerber File Generation

The universal handoff format to fabrication

Gerber & Manufacturing Output Files

.GTL

Top Copper

.GBL

Bottom Copper

.GTS

Top Soldermask

.GBS

Bottom Soldermask

.GTO

Top Silkscreen

.GBO

Bottom Silkscreen

.GKO

Board Edge

.DRL

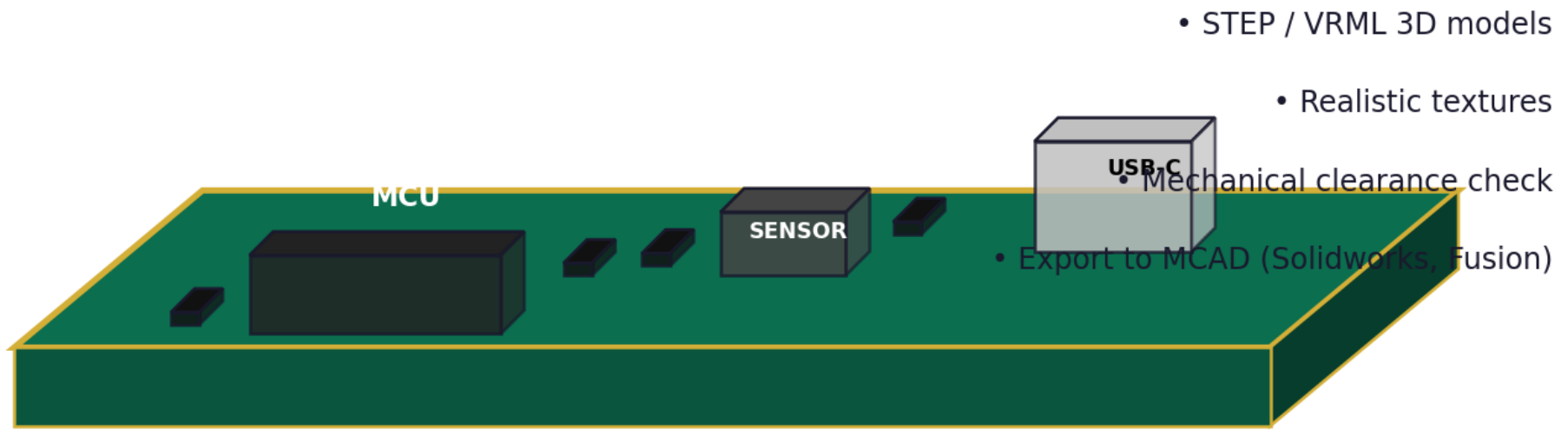
Drill File

Plus a Pick-and-Place (.pos) and BOM (.csv) for SMT assembly

3D PCB Visualization in KiCad

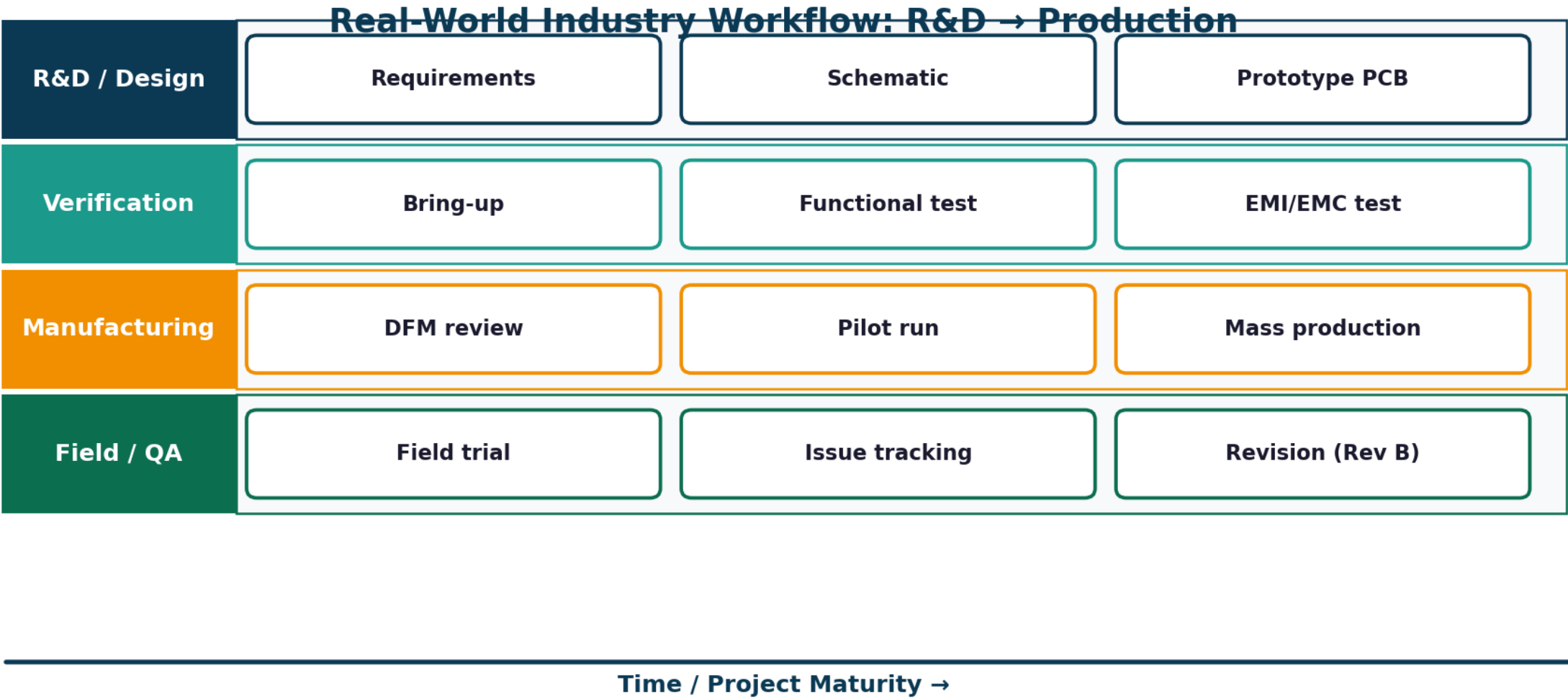
Mechanical fit checks before fabrication

3D Visualization in KiCad — Mechanical Fit Check



Real-world Industry Workflow

From R&D bench to mass production



Common Mistakes in PCB Design

Learn from the failures so you don't repeat them

Common Mistakes — and Why They Hurt



Missing decoupling caps

Voltage spikes, MCU resets



90° trace corners

Acid traps in fab + impedance bump



Split ground planes

EMI, slow returns, noisy signals



No fiducials/test points

Hard to assemble & debug



Tiny silkscreen

Unreadable after fab



Wrong footprint

Board fails at assembly



Skipping DRC

Shorts, opens, scrap



Power traces too thin

Voltage drop, heating

Best Practices

The habits that separate hobbyists from engineers

PCB Design Best Practices

✓ Start with a clean, hierarchical schematic

✓ Place decoupling caps closest to power pins

✓ Use 45° (or curved) traces, never 90°

✓ Add test points & fiducial marks

✓ Verify footprints against component datasheet

✓ Document every revision (Rev A → Rev B with notes)

✓ Use a continuous ground plane on inner layer

✓ Match impedance for high-speed signals

✓ Length-match clock & differential pairs

✓ Keep analog & digital grounds separated, joined at one point

✓ Run ERC, DRC, and a manual review every revision

✓ Build a 3D model for mechanical fit early

Future Trends in PCB Design

Where the industry is heading

Future Trends in PCB Design

1 HDI / Microvia

$\leq 100 \mu\text{m}$ features
for mobile & wearables

2 Flex / Rigid-Flex

Foldable, IoT, medical implants

3 Embedded Components

R/C inside layers — denser boards

4 AI-Assisted Routing

ML-driven autorouters & placement

5 IPC-2581 / ODB++

Open formats replacing Gerber

6 Sustainable PCBs

Halogen-free, recyclable substrates

Conclusion

Key takeaways from today's session

Master the workflow

Schematic → ERC → Layout → DRC → Gerber
Iterate, don't ship Rev A.

KiCad is industry-ready

Free, open-source, Python-scriptable, used by
startups and Fortune-500 alike.

Placement before routing

60% placement, 40% routing.
Good placement = signal integrity for free.

Ground plane = EMI insurance

Continuous reference plane on a 4-layer board
solves most EMC headaches.

Footprints = source of truth

Verify every footprint against the datasheet.
It's the #1 cause of re-spins.

Practice on real boards

Design, fab, bring up, debug.
Three boards in your portfolio beats any certificate.

Questions & Discussion

Ask anything — design choices, KiCad workflow, industry practices, project ideas.

References & Further Learning

- ▶ [kicad.org](https://www.kicad.org) — official downloads, docs, and community forum
- ▶ docs.kicad.org — getting-started, tutorials, scripting reference
- ▶ snapeda.com • componentsearchengine.com — vetted symbols & footprints
- ▶ Phil's Lab (YouTube) — high-quality KiCad and signal-integrity walkthroughs
- ▶ Robert Feranec (YouTube) — advanced PCB & SI courses
- ▶ Henry Ott — Electromagnetic Compatibility Engineering (definitive EMC reference)
- ▶ Howard Johnson — High-Speed Digital Design (signal integrity classic)
- ▶ IPC-2221 / IPC-7351 — industry standards for PCB & footprint design

PCB Manufacturing Process

Single-Layer • Double-Layer • Multilayer

From copper-clad laminate to finished circuit board

Audience: Engineering Students & Junior Engineers

Materials • Imaging • Etching • Lamination • Drilling • Plating • Mask • Finish • Test

Why Understand the Manufacturing Process?

Design decisions you can't make without it

Cost

Every design choice — layer count, via type, finish — changes price.

Yield

Designs that fight the process lose 5-30% yield.

Lead time

Multilayer adds 1-2 weeks vs double-layer turnaround.

DFM

You can't design for manufacturability if you don't know the steps.

Debugging

Spot defects → name them → fix them in the next design iteration.

Career

Hardware engineers who get this are 10× more useful in industry.

What is a PCB?

Substrate + copper + protective layers

- ▶ Printed Circuit Board — substrate carrying copper conductors
- ▶ Replaces hand-wired point-to-point circuits
- ▶ Three core ingredients: substrate, copper, protection (mask + silkscreen)
- ▶ Substrate is usually FR-4 (Fibreglass-Reinforced epoxy, type 4)
- ▶ Copper layers carry signals + power + ground
- ▶ Solder mask protects copper, defines pads
- ▶ Silkscreen labels components for assembly

Key dimensions

Typical 2-layer FR-4 board: 1.6 mm thick
Standard copper weight: 1 oz/ft² (35 µm thick)
Cu trace width: 6-20 mil (0.15-0.5 mm) typical
Drill holes: 0.3-3 mm

Industry naming

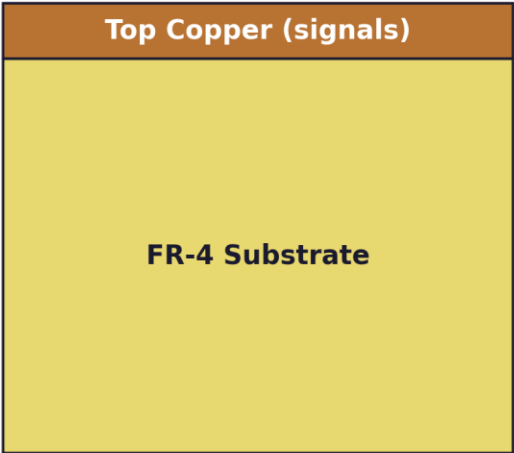
Bare board / Bare PCB → just the substrate + copper (no parts)
PCBA → PCB Assembly (board with components soldered on)
Fab + Assembly → 'turnkey' service from a CM

PCB Types — By Layer Count

Single-layer / Double-layer / Multilayer

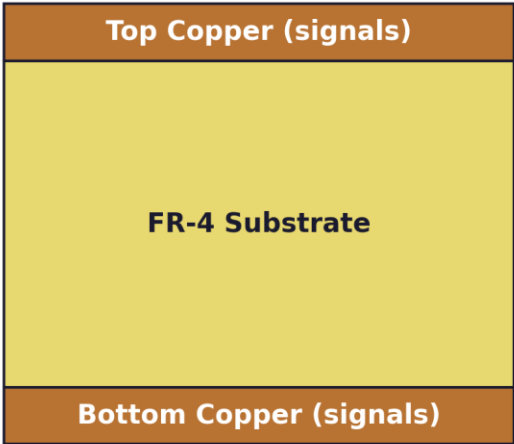
PCB Types — Cross-Section Comparison

Single-layer



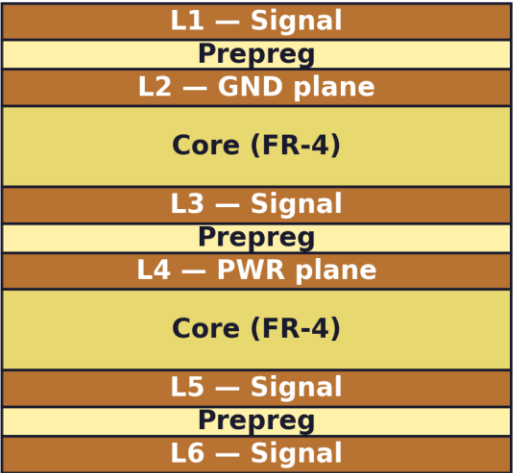
1 copper layer • cheapest • toys, kits, low-frequency

Double-layer



2 copper layers • most popular • general-purpose

Multilayer (6-layer shown)



4 / 6 / 8+ layers • SI / EMC / density

Single-Layer PCB

1 copper layer • cheapest tier • limited applications

- ▶ Copper foil on one side of the substrate only
- ▶ All components and traces on the same side
- ▶ No through-hole plating (PTH not needed — no second side)
- ▶ Cheapest PCB type — ~₹100-300 per board at low volume
- ▶ Simple process: laminate → etch → solder mask → silkscreen → cut
- ▶ Limitations: low density, no crossings (need jumpers), low frequency

Typical applications

- LED-driver boards
- Power-supply boards (AC-DC adapter PCBs)
- Toy / kit boards
- Calculator / remote-control PCBs
- Cost-sensitive consumer electronics

When NOT to use

Any design with:

- Multiple ICs that need to interconnect freely
- High-speed signals (clocks, USB, Ethernet)
- Mixed analog + digital sections
- Components on both sides

Double-Layer PCB

2 copper layers • Plated through-holes • Workhorse of the industry

- ▶ Copper foil on both top and bottom of the substrate
- ▶ Plated through-holes (PTH) connect top ↔ bottom layers
- ▶ Components can be placed on both sides (SMD top, THT bottom etc.)
- ▶ Most common PCB type for prototypes & consumer products
- ▶ Cost: ~₹500-1500 for 50×50 mm at low volume
- ▶ Lead time: 3-5 days at fast fabs (JLCPCB / PCBWay)

Typical applications

- Arduino / RPi / ESP32-style hobby boards
- Industrial sensor modules
- Small-batch products
- Education / prototype work
- Most commercial products without high-speed digital

Limitations

- No dedicated power / ground plane (shared GND on bottom)
- EMC harder than 4-layer
- Cross-talk between traces above ~50 MHz
- Need careful manual routing — no internal layers to help

Multilayer PCB

4 / 6 / 8+ copper layers • Dedicated power/ground • SI + EMC king

- ▶ Multiple cores (FR-4 with copper) bonded with prepreg
- ▶ Inner copper layers dedicated to power and ground planes
- ▶ Outer layers for signals — every signal sees a continuous reference plane
- ▶ 4-layer is the modern baseline for non-trivial products
- ▶ 6 / 8 / 12 layer for dense MCUs, FPGAs, mixed-signal
- ▶ BGA packages with > 200 balls usually require ≥ 6 layers

Why more layers?

1. Dedicated ground plane = clean returns + EMC
2. Dedicated power plane = low-impedance distribution
3. Inner signal layers free up outer-layer space
4. Controlled impedance for high-speed traces
5. Higher routing density (BGA escape)

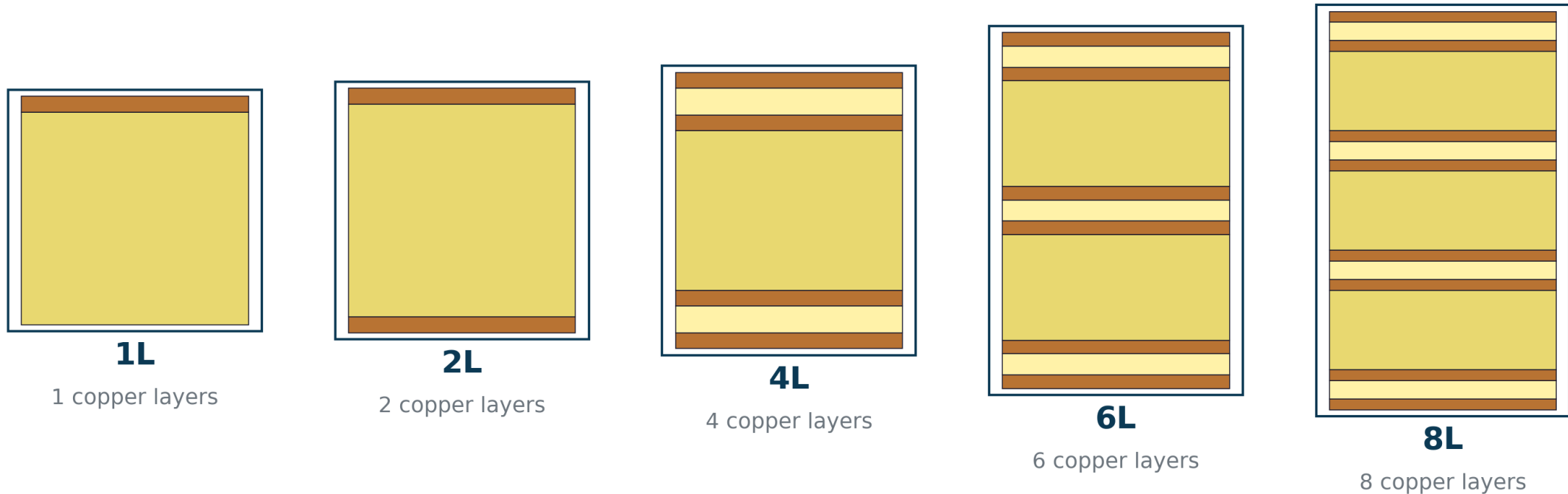
Cost & lead-time impact

4-layer: 2-3× the cost of 2-layer at low volume
6-layer: 4-5×
Lead time: +3-5 days for lamination & extra steps
Often worth it — SI / EMC pain savings > extra cost

Stackup Comparison

1 / 2 / 4 / 6 / 8 layer cross-sections (to scale)

Stackup Comparison — 1 / 2 / 4 / 6 / 8 Layers (to scale)



*More layers → better signal/power integrity, denser routing, higher cost.
4-layer is the modern industry baseline for non-trivial products.*

Manufacturing Flow — 13 Stages

Single & double-layer skip the multilayer-specific stages (2-4)

PCB Manufacturing Process — 13-Stage Flow



Stages 2-4 apply ONLY to multilayer boards. Single & double-layer skip inner-layer + lamination.

Step 1 — Materials Prep

Substrates, prepregs, copper foils, surface treatments

Step 1 — Materials & Substrates

FR-4 Core

Glass-epoxy laminate
0.4–1.6 mm typical
 $Dk \approx 4.2\text{--}4.7$

Prepreg

B-stage glass cloth +
uncured resin
Bonds layers in pressing

Copper foil

Electrodeposited copper
 $\frac{1}{2}\text{oz}$ / 1oz / 2oz / heavy
(17 / 35 / 70 μm)

Resin systems

Epoxy (FR-4) most common
Polyimide (Kapton) for flex
Rogers for RF

Reinforcement

E-glass fibre weave
or non-woven aramid
(driving $Dk + Tg$)

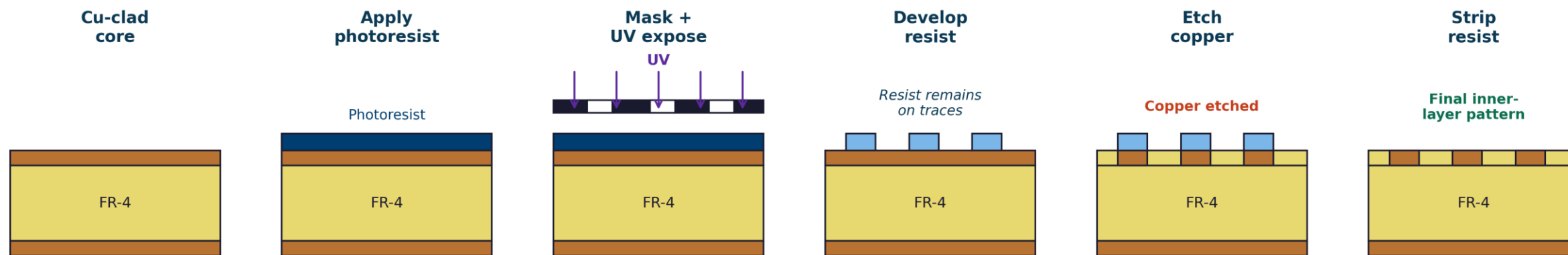
Surface treatments

Black/brown-oxide treatment
on Cu for prepreg adhesion

Steps 2-3 — Inner-Layer Imaging + Etching

Multilayer ONLY • Photolithography pattern transfer

Inner-Layer Imaging & Etching (Multilayer Only)



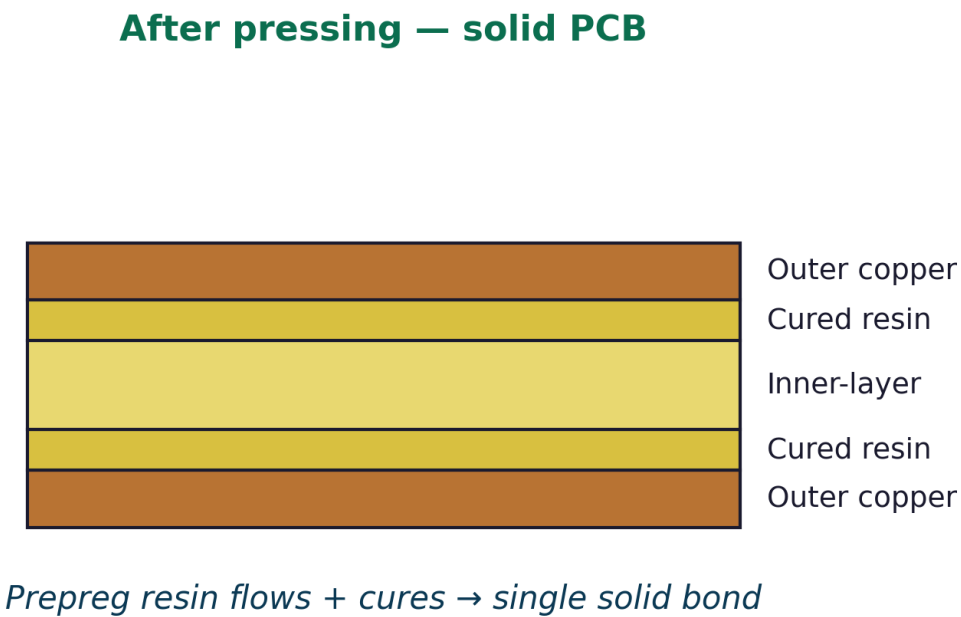
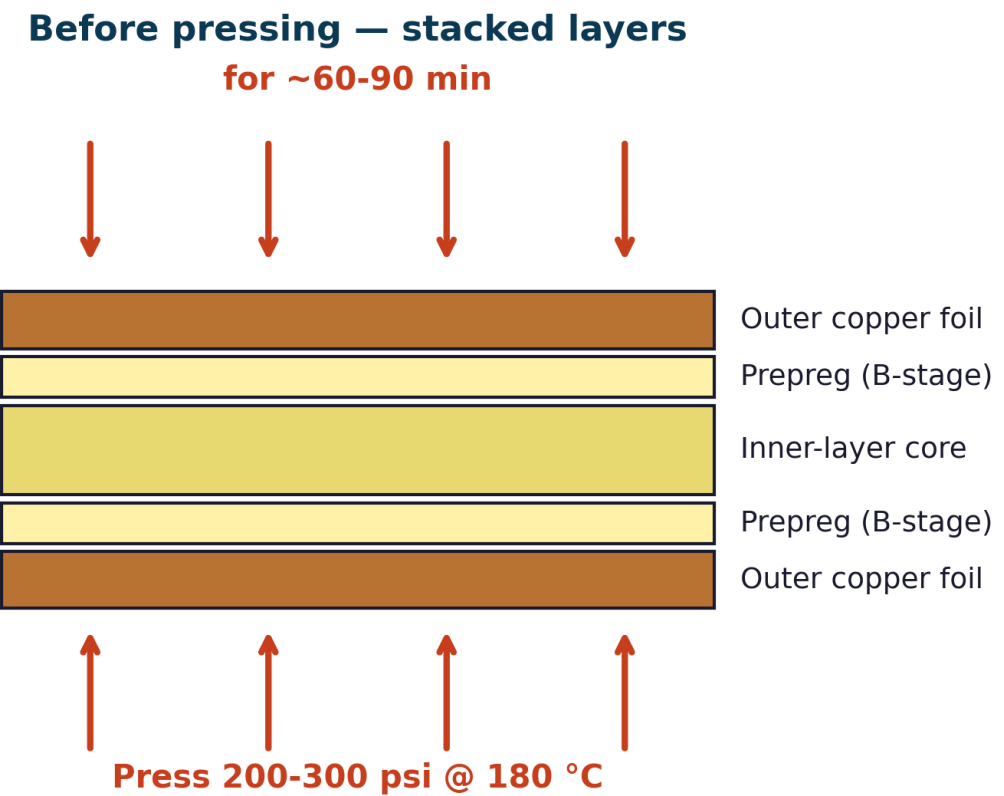
Single & double-layer SKIP this — they have no inner layers to image.

Materials: dry-film photoresist (DFR) laminated to Cu-clad core, UV exposed through phototool, developed in alkaline solution, copper etched in ferric chloride or cupric chloride, resist stripped.

Step 4 — Lamination

Multilayer ONLY • Heat + pressure bonds the stack into one solid PCB

Step 4 — Lamination (Multilayer Only)

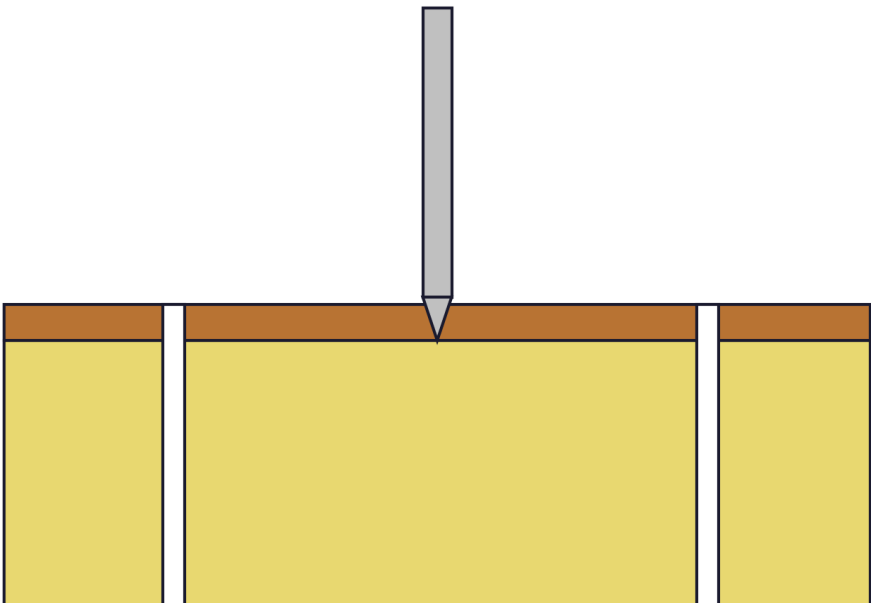


Step 5 — Drilling

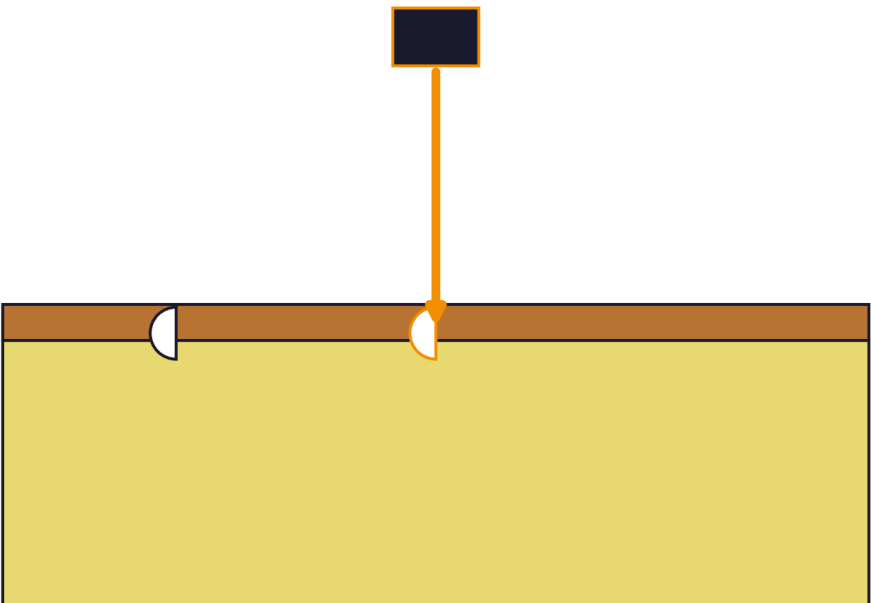
Mechanical drills for through-holes & vias • Laser for microvias (HDI)

Step 5 — Drilling

Mechanical drilling — through-hole & vias
200k rpm



Laser drilling — microvias for HDI
CO₂ / UV Laser



Drilling specs

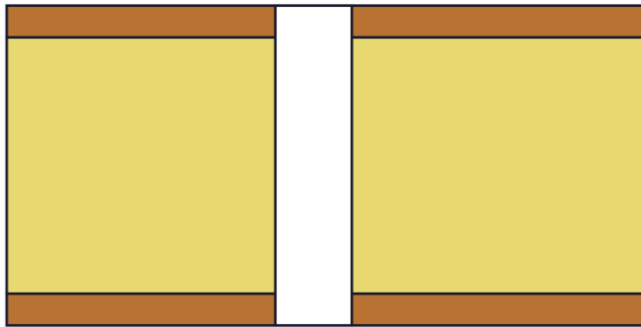
Standard mechanical: 0.3 mm minimum drill, 0.15 mm annular ring • Laser (UV/CO₂): 50-150 µm microvias for HDI • Aspect ratio (depth/diameter) ≤ 10:1 for reliable plating

Step 6 — Plated Through-Hole (PTH)

Make non-conductive hole walls conductive • Two-step plating

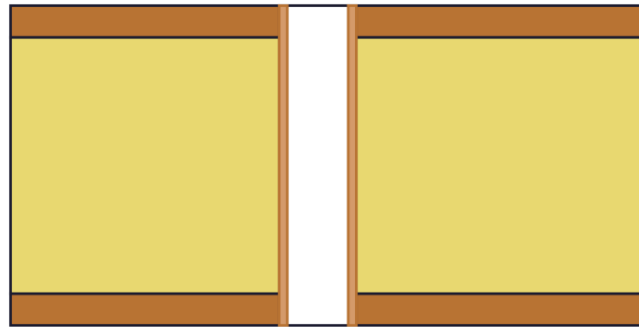
Step 6 — Plated Through-Hole (PTH) Process

(a) After drilling



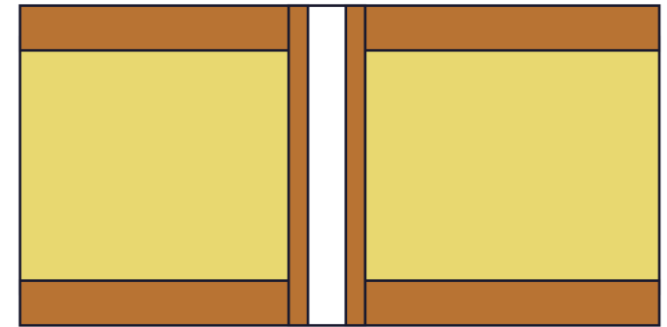
Hole walls = bare FR-4

(b) Electroless Cu seed



Chemical bath deposits

(c) Electrolytic Cu plating



DC current builds Cu to

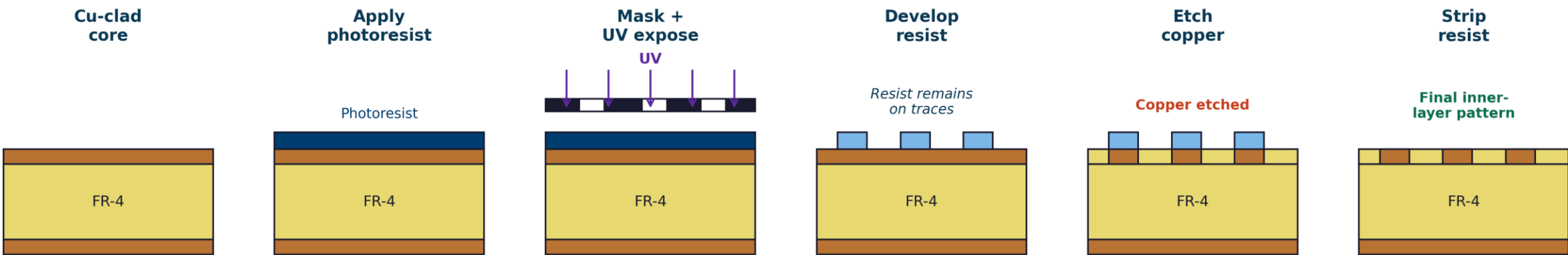
Why two steps?

(a) Electroless copper deposits $\sim 1 \mu\text{m}$ seed everywhere — including non-conductive hole walls. (b) Electrolytic copper plating builds up to $\sim 25 \mu\text{m}$ thickness using DC current — needs the seed to conduct.

Steps 7-8 — Outer-Layer Imaging + Etching

Same photolithography as inner layers — now on the outside copper

Inner-Layer Imaging & Etching (Multilayer Only)



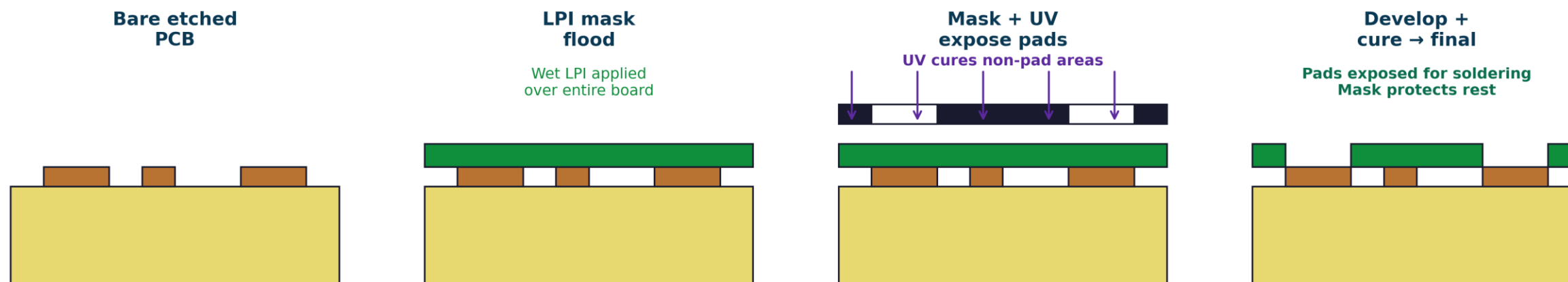
Difference from inner layers

Outer-layer etch must protect PTH holes — uses 'pattern plating' or 'tenting' to prevent the copper plated inside the holes from being etched away. Diagram pattern is the same as steps 2-3, just applied to outer Cu instead of inner.

Step 9 — Solder Mask

LPI (Liquid Photo-Imageable) green/black/red protective layer

Step 9 — Solder Mask (LPI = Liquid Photo-Imageable)



Why solder mask matters

Protects copper from oxidation • Prevents solder bridges during assembly • Defines pad openings precisely • Improves visual readability of silkscreen above • Comes in green (cheapest), black (cosmetic), red, blue, yellow, white, purple

Step 10 — Silkscreen

Reference designators, logos, polarity marks — printed legend

- ▶ Epoxy-based ink applied through a screen or directly by inkjet
- ▶ Default colour: white on green board / black on white board
- ▶ Used for: ref-des (R1, C1, U1), polarity marks, logos, board version
- ▶ Minimum legible text: ~0.8 mm (32 mil) height
- ▶ Avoid silk over pads (gets covered by solder + looks unprofessional)
- ▶ DRC rule: 'silkscreen-on-pad' violation

Modern alternatives

Direct inkjet silkscreen — sharper, faster, more colors
Laser engraving — for high-end, no ink wear
QR codes printed in silkscreen — for serial-number tracking

Cost

Silkscreen is essentially free — included in standard PCB price.
Adding silkscreen on BOTH sides has no real cost impact.
Cosmetic / branding-grade silk is a slight upcharge.

Step 11 — Surface Finish

Protect exposed copper • Make pads solderable

Step 11 — Surface Finishes

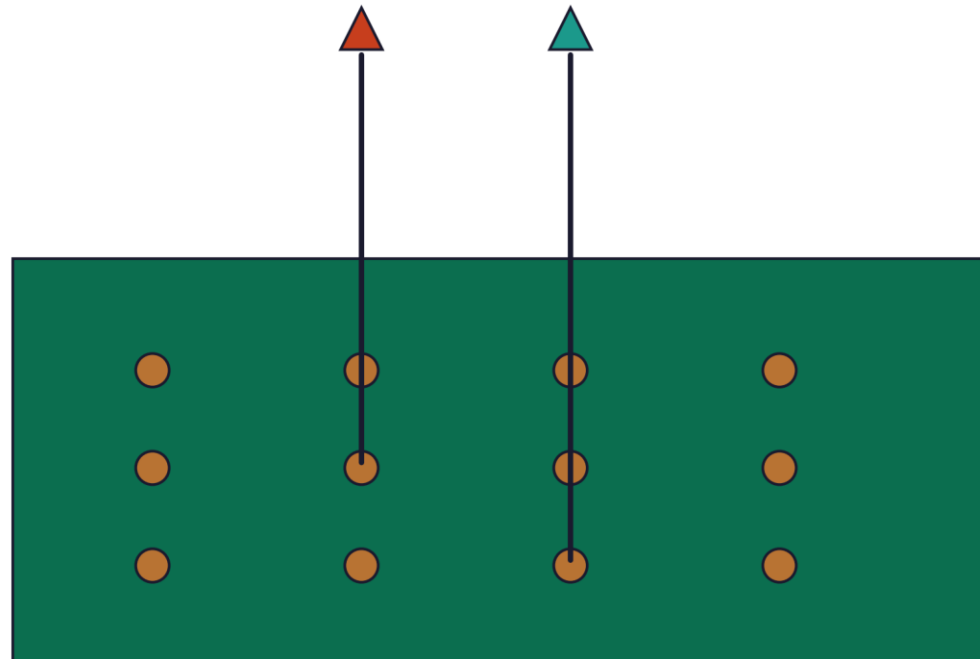
HASL Hot Air Solder Level Sn-Pb dip + air-knife ₹ Cheapest, uneven	Lead-free HASL SAC305 alloy RoHS-friendly ₹₹ Uneven, hot process	ENIG Electroless Ni + Immersion Au Gold finish ₹₹₹ Flat, best for BGA
OSP Organic Solderability Preservative Clear film ₹ Flat, short shelf life	Immersion Ag Pure Ag on Cu Flat finish ₹₹ Flat, tarnishes	Hard Gold Electrolytic Au Thick wear-resistant ₹₹₹₹ For contact pads

Step 12 — Electrical Test

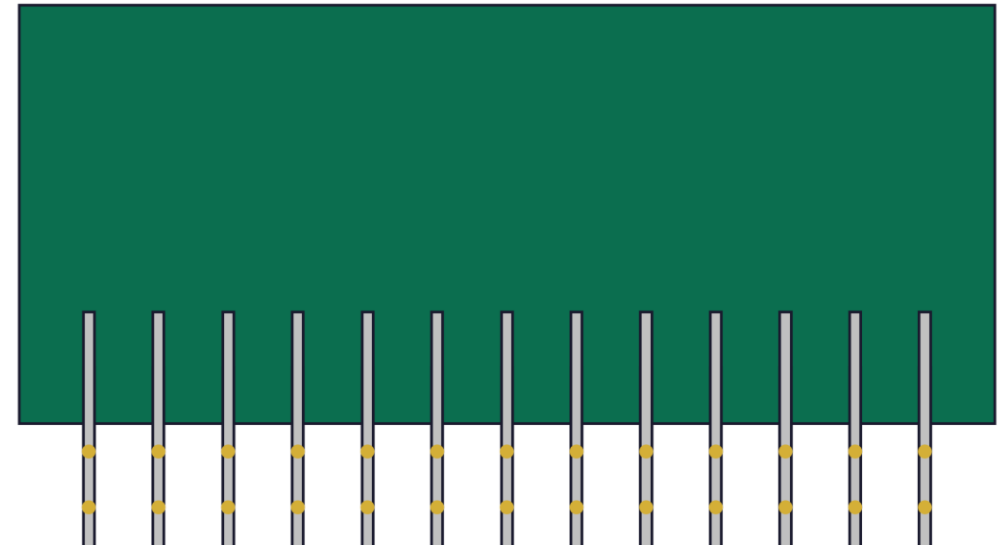
Flying probe (low-vol) vs Bed-of-nails (high-vol)

Step 12 — Electrical Test

Flying probe test



Bed-of-nails fixture



What's being tested?

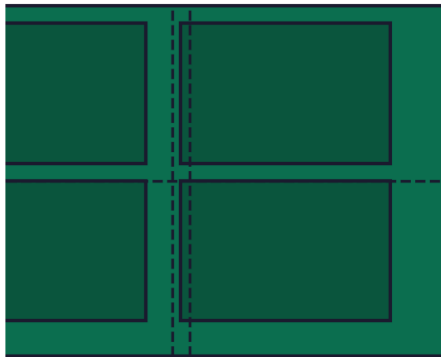
Continuity — every net has zero resistance from end to end • Isolation — adjacent nets have infinite resistance • Hi-pot — no shorts under 500-1500 V DC stress • Result: PASS or FAIL per board; fabs reject failed boards before shipping

Step 13 — Routing & Panel Separation

Cutting individual boards from the manufacturing panel

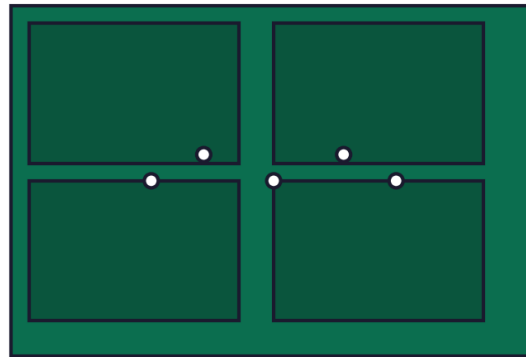
Step 13 — Routing & Panel Separation

V-groove



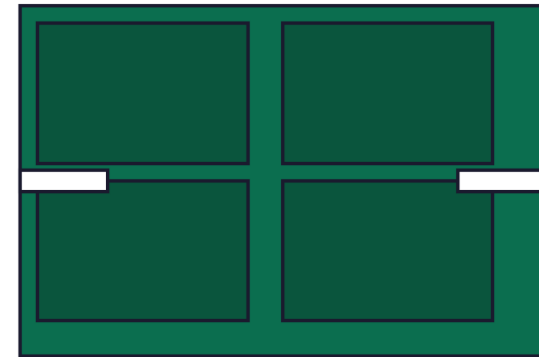
V-cut grooves on top + bottom
Snap apart after assembly

Mouse-bite tabs



Small drilled tabs hold boards together
Break off after assembly

Routed slots + tabs



CNC slot defines outline
Tabs hold panel together

Final step before shipping. Boards may also be packaged in panels for SMT line assembly first.

Process Comparison

What's different between single / double / multilayer

Process Comparison — Single vs Double vs Multilayer

Stage	Single-layer	Double-layer	Multilayer
1. Materials	Cu-clad FR-4	Cu-clad FR-4	Cores + prepreg + Cu foils
2. Inner-layer imaging	X Skipped	X Skipped	✓ Required
3. Inner-layer etch	X Skipped	X Skipped	✓ Required
4. Lamination	X Skipped	X Skipped	✓ Required
5. Drilling	✓ Mech.	✓ Mech.	✓ Mech. + laser (HDI)
6. Plating	X No PTH	✓ PTH required	✓ PTH required
7-8. Outer image+etch	✓ One side only	✓ Both sides	✓ Both outer sides
9. Solder mask	✓ One side	✓ Both sides	✓ Both sides
10. Silkscreen	✓	✓	✓
11. Surface finish	✓	✓	✓
12. Electrical test	Simple	Net-list	Net-list (extensive)
13. Cutting	✓	✓	✓
Lead time	1-3 days	3-5 days	5-15 days

Common Manufacturing Defects

What to look for at incoming inspection

Common PCB Manufacturing Defects

⚠ Open trace

Cu missing → no connection

⚠ Short circuit

Adjacent traces bridged

⚠ Drill breakout

Hole drilled off-pad

⚠ Annular ring

Insufficient pad around drill

⚠ Solder mask misalign

Mask offset → exposed copper

⚠ Lifted pad

Pad detached from substrate

⚠ Cu void

Plating incomplete in PTH

⚠ Layer mis-register

Inner layers offset from outer

How to Specify a PCB Order

Every line on the fab quote form, explained

- ▶ Quantity: 5 / 10 / 50 / 100+ — drives per-unit cost
- ▶ Board size: e.g., 50 × 30 mm — affects panel utilisation
- ▶ Layer count: 1 / 2 / 4 / 6 / 8 — biggest cost driver
- ▶ Material: FR-4 standard (Tg 130 / 170 / 180)
- ▶ Thickness: 0.8 / 1.0 / 1.6 / 2.0 mm
- ▶ Outer copper weight: 1 oz (default) / 2 oz (heavy current)
- ▶ Solder mask color: green (default) / black / red / blue / white
- ▶ Silkscreen color: white (default) / black
- ▶ Surface finish: HASL / lead-free HASL / ENIG / OSP
- ▶ Minimum trace width: 6 mil (default) / 4 mil (premium)
- ▶ Minimum drill: 0.3 mm (default) / 0.15 mm (premium / HDI)
- ▶ Impedance control: Y/N (additional cost & spec sheet)
- ▶ Edge plating / castellated holes: Y/N
- ▶ Panelization: per fab template OR custom panel file

Choosing the Right PCB Type

Decision matrix by criterion

Choosing the Right PCB Type

Criterion	Single	Double	4-layer	6-layer+
Component density	Low	Medium	High	Very high
Signal speeds	DC, kHz	< 50 MHz	< 500 MHz	GHz
Power planes	X	Shared	Dedicated GND	Dedicated GND + PWR
Signal integrity	Poor	OK	Good	Excellent
EMC performance	Poor	Fair	Good	Excellent
Cost (10 pcs, 50×50 mm)	₹500	₹1500	₹5000	₹15k+
Lead time (proto)	1-3 d	3-5 d	5-7 d	10-15 d
Typical use	Toys, kits	General-purpose	MCU + RF	HDI / dense BGAs

Where PCBs Are Made

Global landscape • Indian fabs gaining ground

China

JLCPCB, PCBWay, AllPCB, Seeed
5-7 day lead, very cheap
80% of global PCB volume

Taiwan

Unimicron, Compeq, Zhen Ding
High-end (HDI, IC substrates)
Fastest tech advancement

Korea

Samsung Electro-Mechanics, LG
HDI, flex, advanced packaging

Japan

Ibiden, Nippon Mektron
Ultra-high-end (substrate-like)

USA / EU

Sanmina, Royal Circuit
Higher cost — for ITAR, medical, aero, low-volume

India

Centum, AT&S India, Genus, Epitome
Growing under PLI scheme
Good for proto / small-medium volume

Cost & Lead-Time Guide

What to expect from low-volume prototypes (5-50 pieces)

Spec	Single-layer	Double-layer	4-layer	6-layer
10 pcs, 50×50mm	₹500-1000	₹1000-2000	₹4000-8000	₹15000-25000
50 pcs, 50×50mm	₹1500-2500	₹2500-5000	₹10000-20000	₹30000-60000
Lead time (proto)	1-3 days	3-5 days	5-10 days	10-15 days
Min trace width	8 mil	6 mil	5 mil	4 mil
Min drill	0.4 mm	0.3 mm	0.25 mm	0.2 mm + microvia
Best for	Power, LED	Most products	SI / EMC focus	Dense BGA / RF

Volume break-points

Costs drop steeply between 50 → 500 pcs and again at 5000+ pcs. Real production runs (10k+) are ~10× cheaper per board than prototypes.

Conclusion

Key takeaways from PCB manufacturing

13 stages, predictable

Every PCB follows the same script.
Know the script, design for it.

Layer count drives cost

1L → 2L → 4L → 6L doubles cost each step.
Pick what you need, no more.

Multilayer adds 3 stages

Inner imaging + etch + lamination
are unique to multilayer.

Surface finish matters

HASL for cheap, ENIG for fine pitch, OSP for cost + flat.

Tooling is the cost driver

Per-board cost drops 10× from
prototype to mass production.

DFM saves rev cycles

Designing within fab capabilities = cheaper boards +
faster turnaround.

Questions & Discussion

Ask anything — process steps, vendor selection, defect identification, cost estimation.

References & Further Reading

- ▶ IPC-6012 — Qualification & Performance Specification for Rigid PCBs
- ▶ IPC-A-600 — Acceptability of Printed Boards (the bible of PCB inspection)
- ▶ IPC-2221 — Generic Standard on Printed Board Design
- ▶ JLCPCB / PCBWay knowledge bases — practical fab-side perspective
- ▶ Sierra Circuits — design + manufacturing how-to articles
- ▶ Phil's Lab YouTube — PCB factory tour + DFM walkthroughs
- ▶ MK Publications — Printed Circuits Handbook (Coombs)
- ▶ Indian PCB Association (IPCA) — local fab capabilities directory

Component Packages

Resistors, Capacitors, Inductors & IC Packages

How to read a datasheet — and pick the right footprint

Audience: Beginner → Intermediate Engineers

Through-hole • SMD • Leadless • BGA • Selection criteria • Assembly

Why Component Packages Matter

Six dimensions in every package decision

Why Component Packages Matter

Footprint

Determines PCB landpattern and placement density

Assembly

Hand-solder vs reflow, pick-and-place feasibility

Thermal

How heat leaves the die — lead frame, pad, or balls

Electrical

Lead inductance, ESR/ESL, parasitic capacitance

Cost

Larger = cheaper to assemble, smaller = cheaper to ship

Reliability

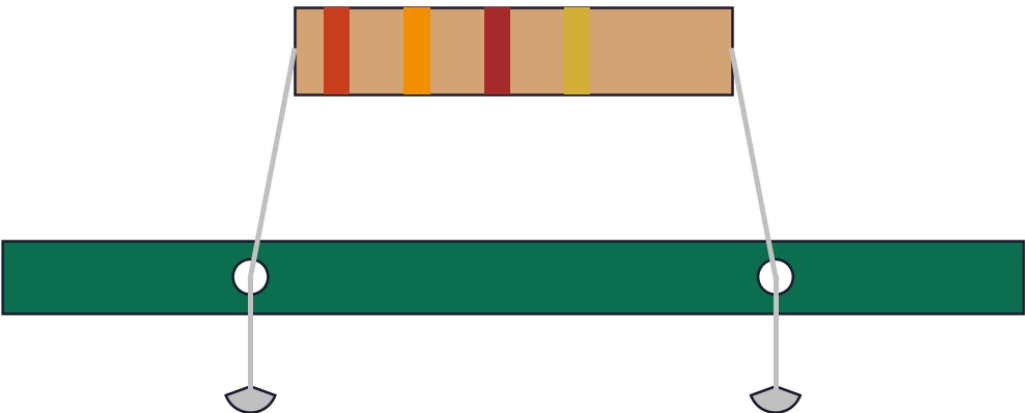
Solder-joint stress, vibration, thermal-cycle endurance

Through-Hole vs Surface-Mount

Two assembly worlds — and where each still wins

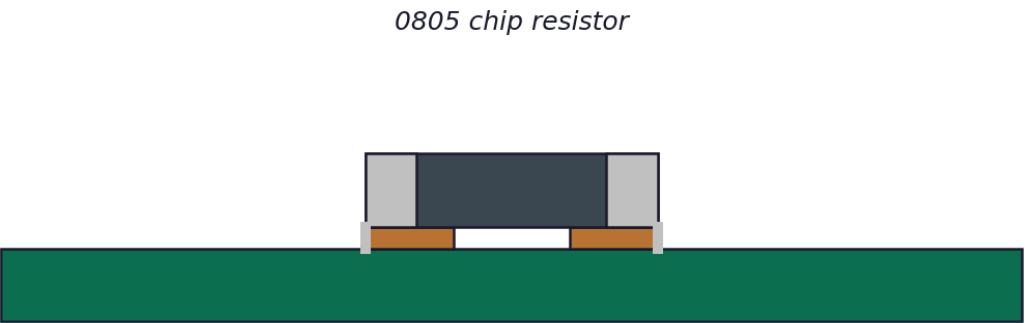
Through-Hole vs Surface-Mount Technology

Through-Hole (THT)



Leads pass through PCB, soldered on the opposite side

Surface-Mount (SMD / SMT)



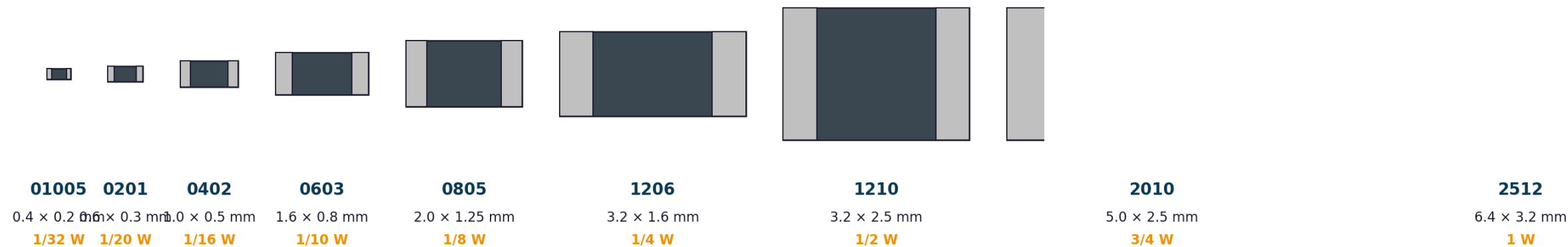
Sits on top of PCB — reflow-soldered onto solder paste

	Through-Hole (THT)	Surface-Mount (SMD)
Mechanical strength	Excellent (lead anchors)	Good (solder only)
Assembly	Wave / hand soldering	Reflow oven, automated SMT
Density	Low	Very high (both sides)
Best for	Connectors, big power, prototyping	Most production: passives, ICs, modules

SMD Chip Sizes — Imperial Codes

0201 / 0402 / 0603 / 0805 / 1206 / 1210 / 2010 / 2512

SMD Chip Sizes (Imperial Codes — drawn to relative scale)



Smaller = denser boards, but harder to hand-solder. 0402 / 0603 dominate consumer & industrial designs; 0201 is mobile-grade.

How to read the code — e.g. '0805' = 0.08" × 0.05" = 2.0 × 1.25 mm

Industrial baseline: 0603/0805. Mobile/wearable: 0402 → 0201. High-power: 1206 → 2512. Metric code exists too (e.g. 0805 = 2012M).

Through-Hole Resistors

Axial bodies, color bands, power dissipation

Through-Hole Resistor Packages



Axial — 1/4 W carbon film

Most common THT resistor



Axial — 1 W metal film

Power dissipation in linear regulators



Wirewound 5 W (ceramic body)

Power supplies, snubbers, dummy loads



TO-220 power resistor

Heatsinkable, 10-50 W

Standard E-series (E12 / E24 / E96) • Tolerances $\pm 5\%$ / $\pm 1\%$ / $\pm 0.1\%$ • TC: ± 50 to ± 100 ppm/ $^{\circ}\text{C}$

Specialty Resistor Packages

Arrays, MELF, current-sense, NTC/PTC, varistors

Specialty Resistor Packages



Resistor Array (R-pack, SIP/MELF) (Metal Electrode Leads)

Bus pull-ups, isolated, common-bus Excellent stability, automotive Shunt resistors, mΩ values, 1–3 W



Current-sense (low Ω, 2512)



NTC / PTC / Varistor

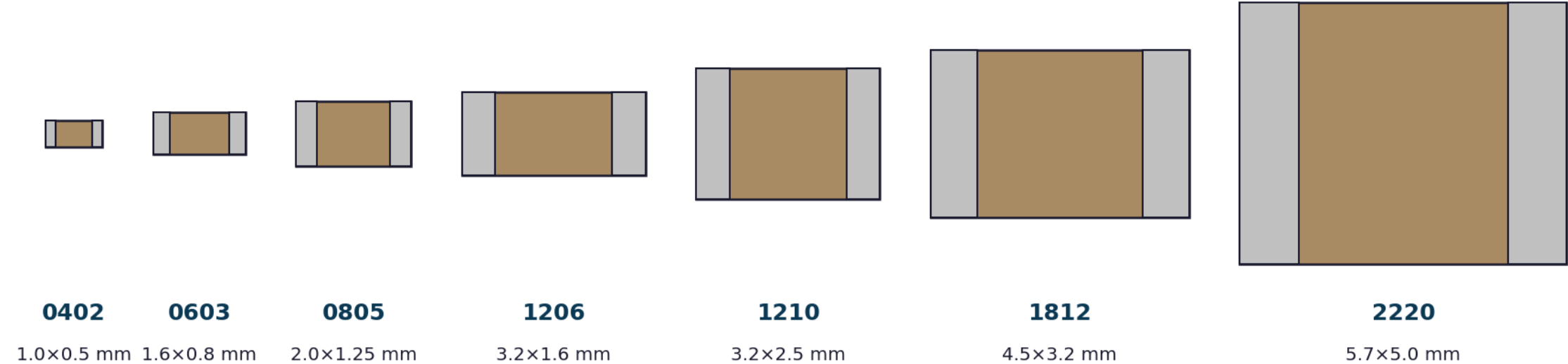
Inrush limiting, surge protection

Tip: pick MELF for high-stability analog, large SMD (2010/2512) for current sensing, and arrays to save BOM lines.

Ceramic (MLCC) Capacitors

The workhorse — 0402 to 2220, COG to Y5V

Ceramic (MLCC) Capacitor Packages



MLCC Dielectric Classes

C0G / NP0: ±30 ppm/°C | ultra-stable, low value (< 100 nF) | filters, timing

X7R: ±15% over -55..+125 °C | general-purpose decoupling, bulk | most common

X5R / X6R: ±15% over -55..+85/105 °C | high-density decoupling

Y5V / Z5U: ±82% over temperature | use only where stability is irrelevant

Tantalum & Polymer Capacitors

Case codes A/B/C/D/E — polarised, low ESR

Tantalum & Polymer Capacitors



A (3216)

3.2×1.6 mm



B (3528)

3.5×2.8 mm



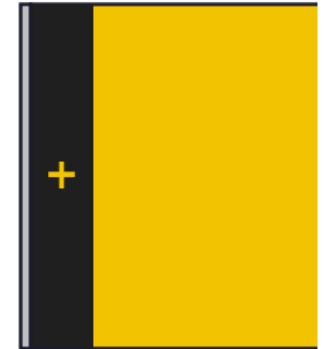
C (6032)

6.0×3.2 mm



D (7343)

7.3×4.3 mm



E (7260)

7.3×6.0 mm

- ▶ Polarised — reverse voltage = failure (sometimes flames!)
- ▶ Choose 2× voltage derating: 3× for polymer tantalum
- ▶ Polymer Ta: lower ESR, safer benign failure mode

▶ Industry alternative to bulky aluminum electrolytics in dense designs

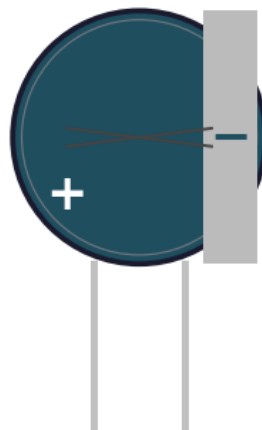
Polarised — orientation matters! • Solid Ta₂O₅ dielectric, MnO₂ or polymer cathode

Polymer tantalum: lower ESR, safer (no ignition failure mode), used in fast-load PSUs

Aluminium Electrolytic Capacitors

Radial / axial / SMD V-chip / snap-in

Aluminium Electrolytic Capacitors



Radial THT

Up to 10 mF, 450 V



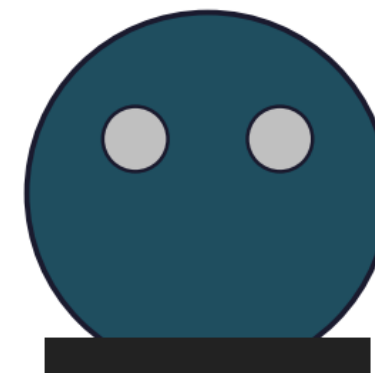
Axial THT

Tube amps, vintage repairs



SMD V-chip

Reflow-compatible alum.



Snap-in / Screw-terminal

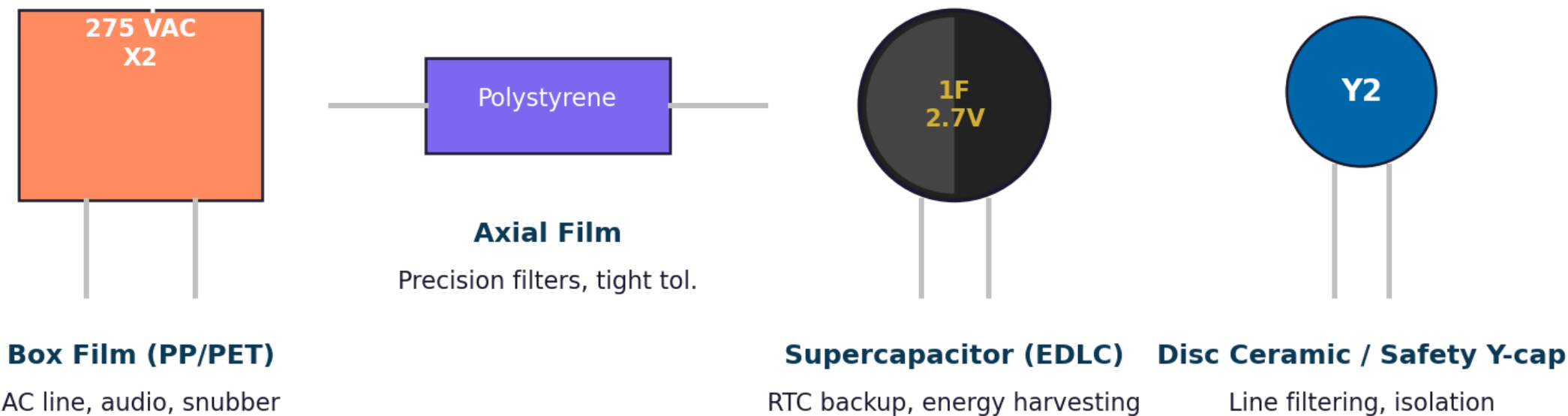
Industrial PSUs, EVs

Polarised • Lifetime halves per +10 °C • Watch ripple-current rating, not just capacitance

Film & Special Capacitors

Polypropylene, polystyrene, supercaps, safety caps

Film & Special Capacitors



Film caps excel in linearity & lifetime; supercaps trade ESR for very large capacitance.

Inductor Packages

Chip, shielded power, toroidal, axial, common-mode chokes

Inductor Packages



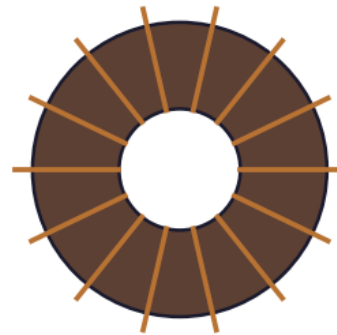
Multilayer Chip (0805/1008)

Ferrite/ceramic, $\leq 100 \mu\text{H}$, RF



Shielded Power Inductor

Buck/boost SMPS, up to 10 A



Toroidal

PFC chokes, line filters



Axial Leaded

Looks like a resistor!



CMC / Transformer

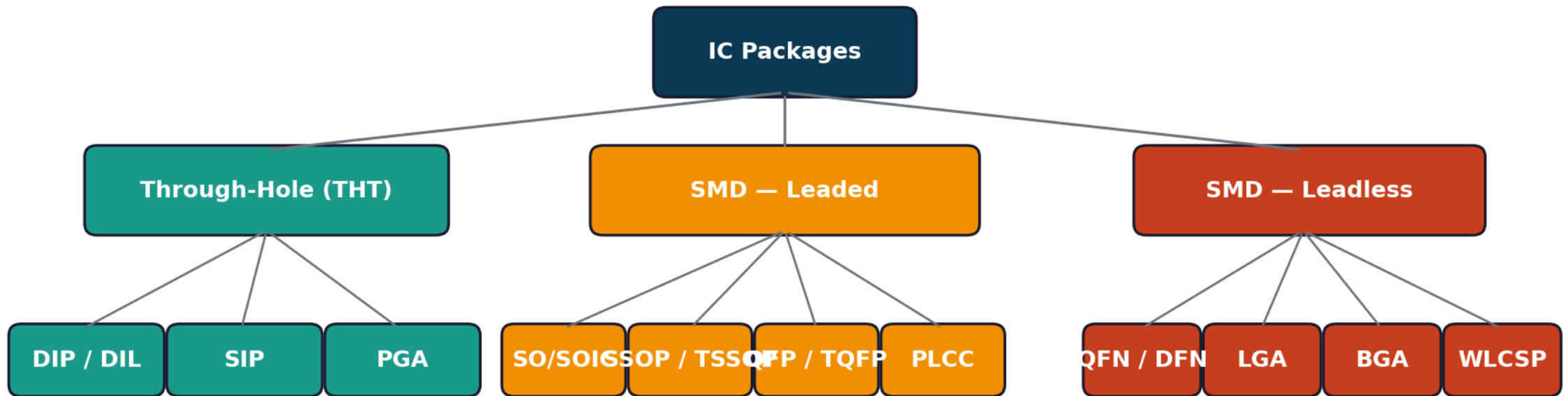
EMI suppression, isolation

Shielded = magnetic field contained (low EMI). Unshielded = cheaper but radiates.

IC Package Family Tree

Through-hole vs SMD-leaded vs SMD-leadless

IC Package Family Tree

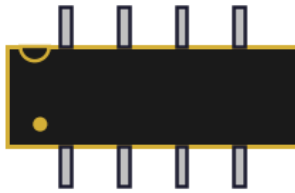


Density & pin count increase → from DIP (≤ 40 pin) to BGA (≥ 1000 pin)

Through-Hole IC: DIP / SIP / PGA / ZIP

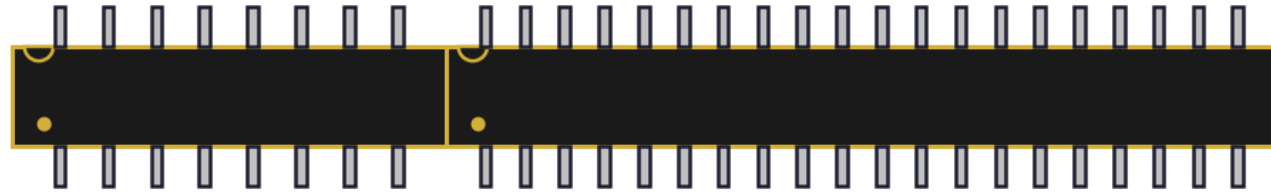
Where you'll still meet plastic dual-inline packages

Through-Hole IC Packages



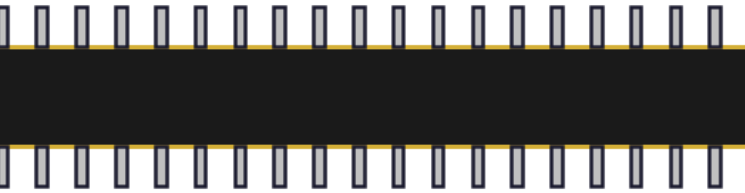
DIP-8 (e.g. 555, op-amps)

2.54 mm pitch, 0.3" wide



DIP-16

74-series logic, EPROM



DIP-40 (ATmega328, 6502 MPU)

0.6" wide variant



SIP (single-inline)

Resistor packs, modules



PGA — Pin Grid Array (legacy CPUs)



ZIP (zigzag inline) — high-density memory

Small-Outline Family: SO / SOIC / SSOP / TSSOP

Gull-wing leaded SMD — easy to inspect, easy to rework

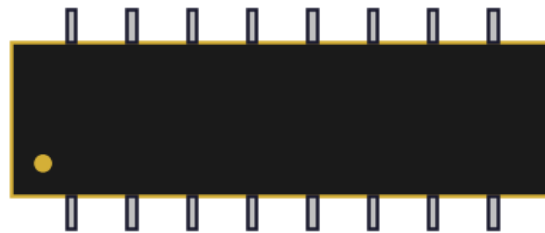
Small-Outline (SO) Family — Gull-wing leaded SMD



SOIC-8 (narrow)

Op-amps, EEPROMs

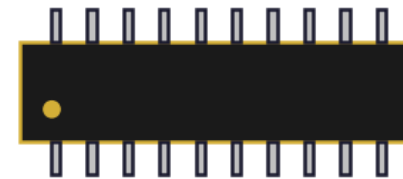
1.27 mm pitch



SOIC-16 (wide)

Logic, interfaces

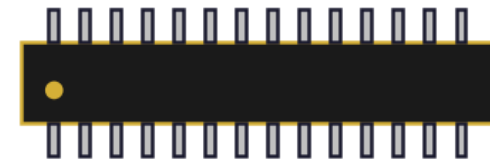
1.27 mm pitch



SSOP-20

Shrunk variant

0.65 mm pitch



TSSOP-28

Thin SSOP, MCUs

0.65 mm pitch

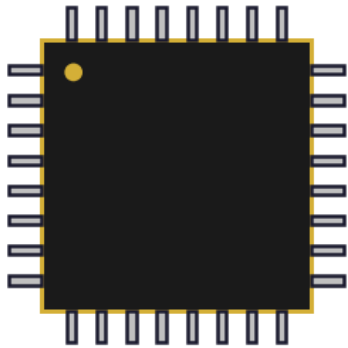
Easiest SMD family to hand-solder

MSOP (mini-SO), SOT-23 (3-lead transistor SMD), TSOP (thin SO for memory) — all share the same gull-wing concept

Quad Flat Pack: QFP / TQFP / LQFP / PLCC

Gull-wing or J-leads on all four sides — pin counts 32 → 256+

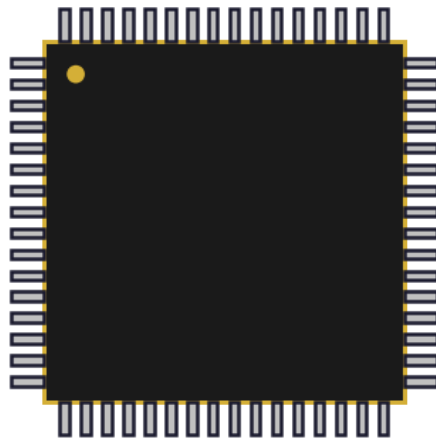
Quad Flat Pack (QFP) Family — leaded on 4 sides



TQFP-32

7×7 mm, 0.8 mm pitch

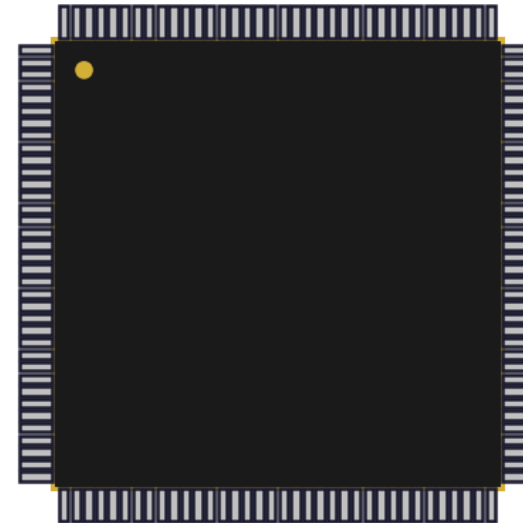
AVR / STM32 entry



LQFP-64

10×10 mm, 0.5 mm pitch

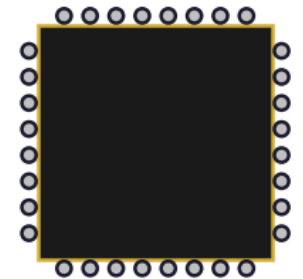
STM32F4 / ESP32 series



LQFP-144

20×20 mm, 0.5 mm pitch

High-end MCUs, FPGAs



PLCC

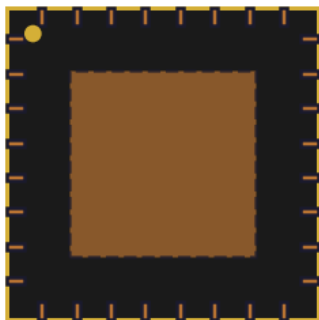
J-leads, socketable
Legacy EEPROMs, BIOS

TQFP = thin, LQFP = low-profile. 0.5 mm pitch is the practical hand-solder limit.

Leadless: QFN, DFN, LGA

Pads underneath — modern default for power-conscious designs

Leadless: QFN, DFN, LGA



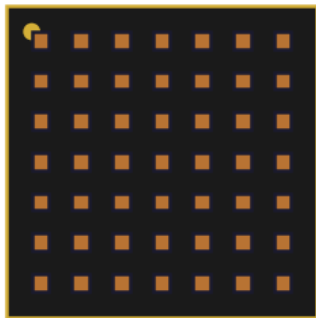
QFN-32

5×5 mm, 0.5 mm pitch
Center pad → thermal/GND



DFN-8 / WSON

2 rows, 0.5 mm pitch
LDOs, MEMS, power switches



LGA (Land Grid Array)

Pads only — no balls
IMUs, RF modules, sockets



Cross-section view

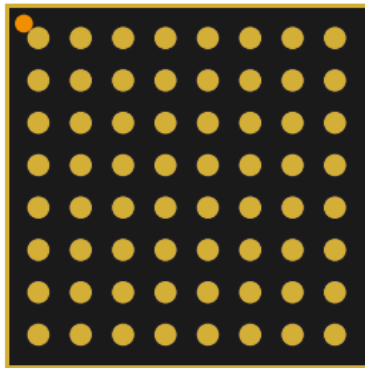
Pads on bottom — no leads to bend

Leadless = lower lead inductance, smaller footprint, but needs proper reflow profile and X-ray inspection.

Ball Grid Array & Wafer-Level CSP

When pin counts cross 200 — and visible joints disappear

Ball Grid Array (BGA) & Wafer-Level CSP



BGA (top view)

Solder balls instead of leads

1.0 / 0.8 / 0.5 / 0.4 mm pitch



BGA cross-section

Solder reflow forms joints

Inner balls = inaccessible (escape routing required)



WLCSP

≤ 0.4 mm pitch, no substrate

Mobile SoCs, ultra-thin devices

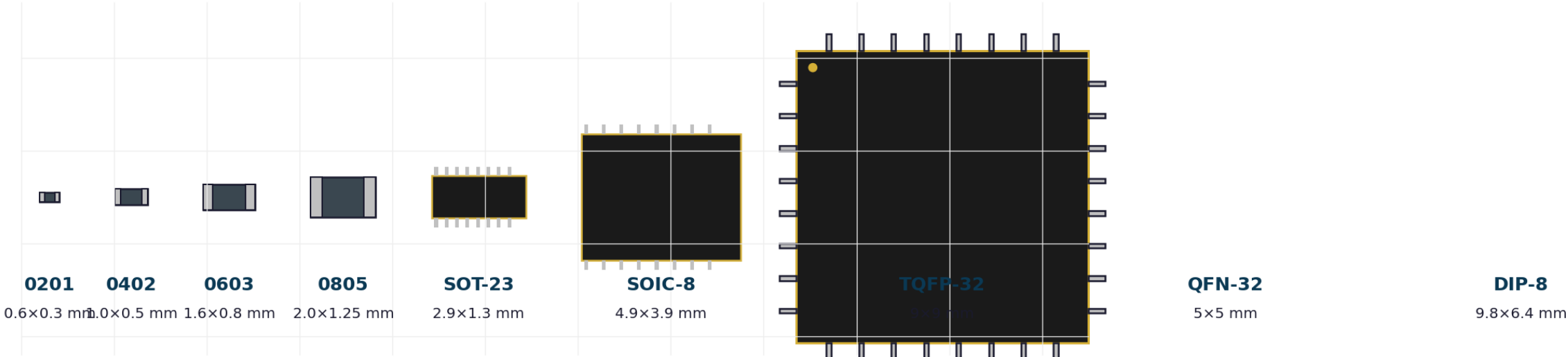
BGA = highest pin density (1000+ pins) — needs ≥ 4 -layer PCB, X-ray inspection, no rework by hand.

Package Size Comparison — Same Scale

From 0201 chip to DIP-40 — drawn at 1 mm = 1 grid unit

Package Size Comparison — Same Scale

(1 grid unit = 1 mm)



From 0.6 mm chip resistor to 10 mm DIP — same boards routinely host the whole range.

How to Choose a Package

Six criteria that drive the decision

How to Choose a Package

Pin Count

More pins → finer pitch / QFN / BGA

Hand-Solder?

0603 / SOIC / TQFP friendly
QFN / BGA need reflow

Board Density

Mobile = WLCSP/BGA;
Industrial = TQFP/QFN

Thermal Load

Look for exposed pad / heatsink tab

Cost & Lead-time

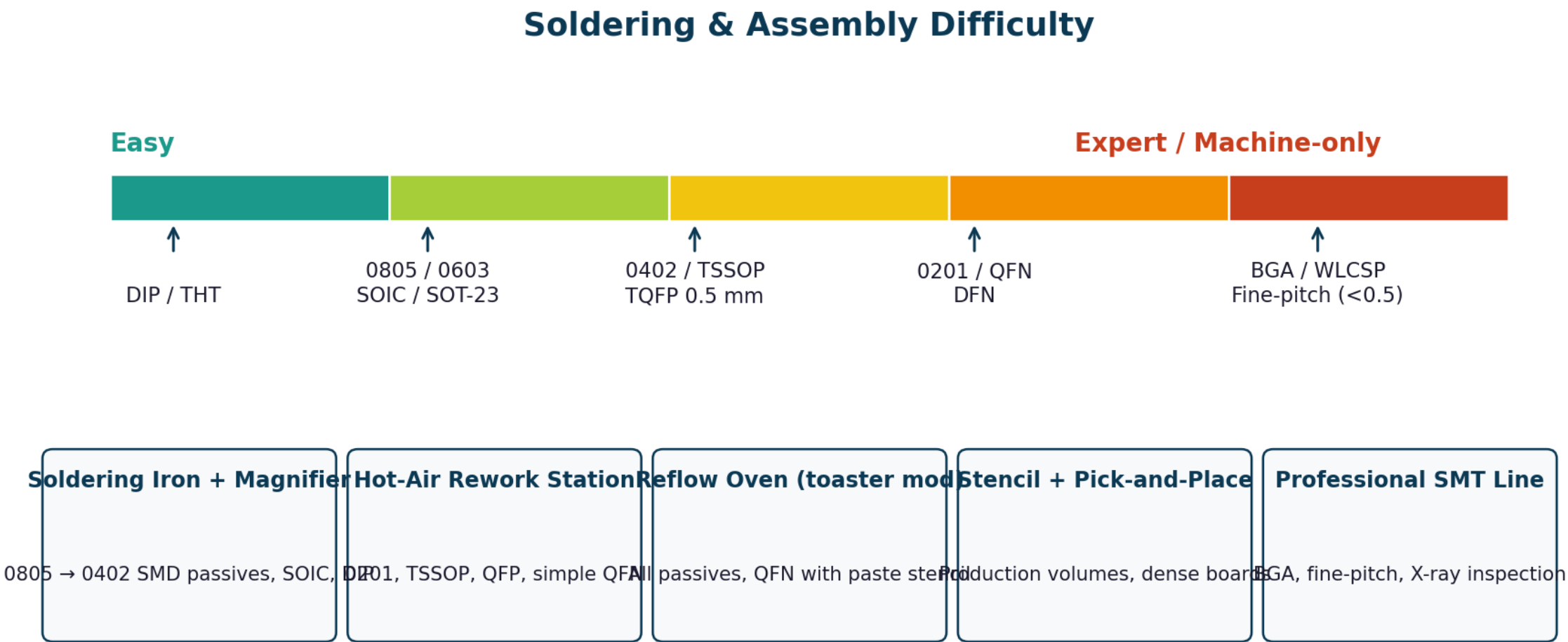
Smaller usually cheaper at volume;
larger easier in prototyping

Reliability / Env.

Automotive / aero prefer leaded
(QFP) over BGA — visual inspection

Soldering & Assembly Difficulty

What you can hand-build vs what needs machines



Quick Reference Cheatsheet

Common packages, pitch, typical pin count, use

Package	Pitch	Pins	Typical use	Hand-solder?
DIP-N	2.54 mm	4–40	MCUs, op-amps, lab/proto	Easy
SOIC	1.27 mm	8–16	Op-amps, EEPROM, drivers	Easy
SSOP / TSSOP	0.65 mm	8–48	Logic, low-pin MCUs	Medium
TQFP / LQFP	0.5 / 0.4 mm	32–256	MCUs (STM32, ESP32), FPGAs	Medium
QFN / DFN	0.5 / 0.4 mm	8–100	LDOs, PMICs, RF, MEMS	Hard
BGA	1.0–0.4 mm	64–2000+	SoCs, FPGAs, RAM, GPUs	Reflow only
WLCSP	≤ 0.4 mm	10–200	Mobile PMICs, sensors	Reflow only
0805 / 0603	—	2	General passives	Easy
0402 / 0201	—	2	Dense / mobile passives	Hard / very hard

Industry Trends in Component Packaging

Where the next 5 years are heading

Where Component Packaging is Heading

1 Miniaturisation

0201 → 01005 → 008004
passives shrinking faster than Moore

2 Integrated Passives

Multi-element arrays,
IPDs replacing 4 discretes

3 Chiplet / 2.5D / 3D

BGA modules with stacked dies
(HBM, Foveros, AMD chiplets)

4 Embedded Components

Resistors/caps inside PCB layers
→ freed surface area

5 Polymer & Hybrid Caps

Replacing aluminium electrolytics
— lower ESR, longer life

6 Lead-Free / RoHS / Halogen-free

Global compliance default

Key Takeaways

Six things to remember from this lecture

Package = Engineering Decision

Footprint, thermal, electrical, assembly, cost — all in one choice.

Default to SMD

Modern boards are SMD-first.
THT only for connectors and heavy power.

Start at 0603 for Passives

Step down to 0402 only if space demands.
Reserve 0201 for production.

Choose Dielectric Wisely

X7R = decoupling. C0G/NP0 = timing/filters.
Never Y5V in a signal path.

Leadless ≠ Lead-free

QFN / BGA hide their solder joints.
Use X-ray or trust your process.

Read the Datasheet — Twice

Verify pitch, body size, thermal pad,
and recommended landpattern.

Questions & Discussion

Ask anything — package choice, soldering, sourcing, datasheets.

References & Further Learning

- ▶ JEDEC.org — official package outline standards (JEP95, etc.)
- ▶ IPC-7351 — generic landpattern standard for SMD components
- ▶ Murata / TDK / Samsung MLCC selector tools — capacitance vs DC-bias data
- ▶ Texas Instruments — package summary tables in every datasheet
- ▶ Eric Bogatin — Signal & Power Integrity Simplified (package parasitics chapter)
- ▶ Phil's Lab (YouTube) — SMD soldering + datasheet reading walkthroughs
- ▶ Application notes from Vishay, Yageo, KEMET on derating and reliability
- ▶ Pick-and-place machine vendor application notes — practical assembly limits

Thank you • Build something this week — even a breakout board.